Slop Ninja

Public detectors, private transformation models, and paid inference on Bittensor

Slop Ninja Research

September 10, 2026
Draft 0.11

Abstract

Slop Ninja proposes a Bittensor subnet launching with two tasks: author and AI-origin detection (A), and author-directed text transformation (B). B jointly targets a particular authorized writer's style and evasion of Pangram and qualified subnet AI-origin detectors. Both objectives guide removal of unwanted generic and formulaic writing while retaining the source's arguments, details and qualifications and matching the requested tonal intent. Indistinguishability from that writer's own work is a target for blinded evaluation, not an achieved result. Every B benchmark evaluates both author fit and detector evasion. Miners publish complete, immutable A inference artifacts and keep B generation models private. They compete on the usefulness of their outputs. They receive network emissions for benchmark performance and can separately sell asynchronous inference at market prices denominated in the subnet's alpha token. Validators execute frozen A artifacts directly; B miners can use them locally for training and search. Pangram remains A's external origin-detection quality anchor, with continuing comparisons against independently documented production histories even after subnet detectors reach parity. Training recipes and source data need not be published merely to reproduce A inference.

Every customer inference request is encrypted for its assigned miner. The customer's source text, reference writing, style profile, and editorial brief are not automatically disclosed to validators, routing services, or other miners. Customers receive encrypted results and may separately provide evidence to a specific validator of their choice. Subnet-owned benchmarks provide the authorized material that validators need to assess accuracy and target style, with preservation, readability, and tone as constraints on B. Transformation miners compete against Pangram and qualified subnet AI-origin detectors. Emission eligibility requires public A artifacts and bounded B rewrite service on authorized A training requests. B miners self-score locally; validators verify those scores without involving A inference endpoints. Retired benchmark rewrites feed later detection rounds. Detector reports establish observations rather than human authorship or preserved meaning.

On-chain contracts coordinate offers, commitments, deadlines, and payments; miners execute inference privately off-chain. Published hash rules determine counterpart and evidence-preparer assignments. Every scored benchmark submission receives the required review at launch. Final judgments require strictly more than two-thirds of frozen eligible validator weight, assuming less than one-third is dishonest and each signer inspects the evidence. This draft specifies the proposed architecture, training data, reward rules, and trust assumptions. It also reports the limits of the existing prototype. Public A artifacts permit reproducible scoring; their accuracy and resistance to behavior that favors allied B models still require independent evaluation. Reliable cleanup in the requested writer's voice, private inference, and live reward settlement remain to be demonstrated.

1 Purpose and scope

Slop Ninja's transformation mechanism jointly targets author-style matching and evasion of Pangram and qualified subnet AI-origin detectors. Both are primary objectives of B. The aim is to remove unwanted generic and mechanically patterned writing while preserving the source's full argumentation, details and qualifications. The result should read naturally in a particular authorized writer's voice, with their characteristic word choices, grammatical constructions and rhetorical habits and the tonal intent of the requested style. Here, *slop* means unwanted filler, vague or formulaic phrasing, needless repetition and other patterns that weaken the intended writing. Intentional informality, unusual constructions and recognizable personal habits belong to the target profile when appropriate to the brief.

The product thesis is that readers object to slop because it combines weak writing with a recognizable generic model voice. Removing those defects should produce good writing that sounds like its author. Successful cleanup therefore means the result has ceased to be slop under this definition. Reader judgments of the writing and author fit must test that thesis alongside the required detector benchmark.

The requested writer's profile supplies a direction for lexical, grammatical and rhetorical changes. Adversarial detector feedback adds pressure to remove patterns recognized as model-generated. B uses both signals to guide the same rewrite. Their practical reinforcement is a testable hypothesis; author fit and detector evasion each retain their own measurements.

The desired endpoint is a revision that readers familiar with the writer cannot reliably distinguish from that writer's own comparable work, while the required substance remains intact and the unwanted patterns have been removed. This is an empirical target. Evaluate it through blinded comparisons in the same register, independent tests on held-out author samples, and full preservation review. A favorable detector score or proximity to a learned profile alone cannot establish that result. A writer may also request greater accessibility or rhetorical force; those changes must respect the source and brief.

The model-publication split is intentional. A should make strong author/origin detection available for anyone to run and use in training, including external detector providers. Matching Pangram-level origin detection is an evaluation target, not an achieved result. Pangram remains A's external origin-detection quality anchor after parity or superiority. Continue comparing both against independent production records; author identification retains its own known-author labels. Public A models give B immediate local feedback while B miners retain their transformation models. A submissions must permit free artifact access, local inference and reuse for training private B models. Hosting still costs compute and may be sold separately. The subnet aims to make detector access a shared resource while miners compete to provide useful private transformation services.

The initial subnet evaluates two tasks separately:

A. Identify authors and documented human/model production histories.
B. Clean up unwanted writing patterns and match an authorized writer's profile while evading Pangram and qualified subnet AI-origin detectors, preserving meaning, detail and target tonal intent.

The detector component of B's strict success target requires the preservation and quality checks, a score strictly below 10% on Pangram's reported AI-generated **plus AI-assisted** fraction, a majority pass against a common panel of three qualified subnet origin detectors, and a pass against the strongest qualified prior-round detector at its calibrated threshold. The panel contains that strongest detector and two hash-selected qualified detectors, frozen for all B candidates in a matched batch (Section 8.2.3). Reaching zero is a desirable measured outcome, not a promise of this design. A model-origin text retains that recorded origin after a successful rewrite. Detector evasion does not change its production history.

The adversarial loop connects these two tasks: detection miners learn to recognize transformed model-origin text, while transformation miners try to defeat qualified detectors from completed rounds. The benchmark fixes the opponent-assignment and scoring rules and retains the source's production records as the models improve.

The initial scope is English prose, document-level author/origin predictions, and bounded asynchronous editing jobs. Span-level origin labels, multilingual rewards, and arbitrary-length manuscripts need their own validated datasets. There is no requirement for token streaming or interactive latency. Writer-directed cleanup belongs to B; there is no separate writing-improvement mechanism at launch. Audience and editorial controls remain part of B's briefs and preservation review.

2 Related subnets, services, and research

This representative survey covers direct task competitors and relevant evaluation and inference infrastructure. It uses project-maintained documentation and research publications checked on September 9–10, 2026. Subnet numbers below follow the projects' own documentation; this is not an independent audit of live chain state. Published service claims are distinguished from results measured by this project.

2.1 Bittensor precedents

It's AI and BitMind's GAS supply the closest task and incentive precedents. It's AI documents human/AI text classification on SN32, with miners serving detection requests. Its FAQ says the team does not know which models its leading miners use. This provides a precedent for evaluating detection endpoints with undisclosed implementations. GAS documents an explicit competition between detection models and generators that try to fool them [1–3].

Project	Documented approach	Relevance to Slop Ninja
It's AI (SN32)	Human/AI text detection, including sentence-level probabilities, through miner endpoints [1,2]	Direct comparison for origin detection; compare calibration and generalization on the same independently labeled text
BitMind GAS (SN34)	Image/video/audio detector submissions and image/video generator endpoints; generation rewards include fooling detectors [3,4]	Direct precedent for adversarial incentives; text revision adds source-preservation and requested-voice constraints
Macrocosmos Finetuning (SN37)	Miners publish competition models on public Hugging Face repositories for validator evaluation [5]	Precedent for rewarding model improvements through submitted weights; Slop Ninja pairs public A artifacts with private B serving
Chutes (SN64)	Serverless model inference and metered service, with documented confidential-serving options [6,7]	Paid inference and private serving infrastructure; writing quality requires separate task evaluation
Targon (SN4)	Managed compute and inference; confidential VMs with attestation-bound key release for proprietary workloads [8,9]	Hardware-based endpoint protection and private-model execution are established design topics

These projects establish that detection competition, adversarial generation, language-model training rewards, and paid private inference already have Bittensor precedents. Their documented mechanisms evaluate different properties. GAS requires submitted detector weights; Slop Ninja also requires public A artifacts, paired with private B generation and independent verification of B's self-scores. Chutes documents confidential serving and request/response encryption, and Targon describes hardware attestation and key release. Those hardware assumptions differ from this paper's default trust in the assigned miner after decryption. A secure execution environment alone does not establish that a revision preserved an argument.

2.2 Existing writing services

Pangram and GPTZero already offer human/model-origin classification, including mixed or assisted categories. Pangram is the required external opponent here; GPTZero is another useful comparison on rights-cleared benchmark material. Their output categories and scores must be mapped explicitly to the evaluation's documented production labels. Agreement among detectors does not establish an author's identity [10,11].

Grammarly documents personal voice profiles and a feature for rewriting in the user's voice. Wordtune offers sentence and paragraph rewriting, formality changes, shortening, and expansion. QuillBot documents multiple paraphrasing modes, including custom and Humanizer modes, and warns that its Creative mode can alter meaning. These are direct comparators for author-directed revision and its editorial controls; personalization and editorial controls are existing product capabilities [12–14].

Undetectable AI explicitly markets humanization modes intended to bypass detectors. That documents an existing commercial objective; this survey does not independently establish its bypass or preservation performance [15]. Comparing such services requires the same sources, briefs, retained outputs, and blinded preservation review used for subnet miners. A product's advertised detector score cannot substitute for that comparison. Hosted service comparisons use authorized benchmark text and do not create an exception to confidential customer delivery.

2.3 Adoption and demand

Model providers and publishing platforms are bringing provenance checks into ordinary writing workflows. Anthropic announced text watermarking on August 14, 2026. Its coverage documentation says Claude models launched on or after August 2 receive marking at launch, while support for older models remains in rollout. Supported outputs are marked worldwide across Claude products, the API and partner platforms [16,17]. Google announced SynthID text watermarking in the Gemini app and web experience in May 2024 [18]. Substack now offers Pangram-powered reader scans for posts and notes published on or after July 21, 2026, with author controls to disable scanning on individual items [19].

These deployments support a market thesis for both subnet tasks. Publishers and readers may want affordable, independently evaluated detection as provenance checks become more common. Writers using model assistance may want stronger control over their own voice and a way to remove unwanted generic phrasing without losing substance. Public A models support accessible detection and local evaluation; private B services compete on the resulting revisions. We expect broader provenance checking to increase demand for these capabilities. The size of that increase, willingness to pay and repeat use remain to be measured through customer pilots.

Anthropic distinguishes keyed watermark detection from classifiers such as Pangram that infer origin from linguistic patterns. Its watermark detector is in private preview for eligible organizations [16]. Passing Pangram and public A models therefore does not establish removal of a Claude watermark. Watermark detectability, inferred origin, prose quality and author fit require separate measurements. The proposed B product remains cleanup in the requested writer's voice; any future watermark benchmark would need access to the relevant detector and its own evaluation protocol.

2.4 Research foundations and proposed contribution

Gero et al. model style through controls over function-word frequencies and syntactic constructions while retaining content words. This is a direct antecedent for combining lexical and grammatical features. StyleRemix uses interpretable style axes and LoRA modules for authorship obfuscation, providing a relevant comparison for controllable rewriting [20,21]. Authorship obfuscation changes evidence available to an author classifier; matching a requested writer and preserving every required claim remain separate evaluation questions.

The PAN/ELOQUENT 2024 task paired human/AI detection with generation and obfuscation intended to avoid it, offering a prior benchmark for an adversarial loop. DIPPER demonstrated paraphrase attacks on several then-current detectors and studied a retrieval-based defense. Those results motivate adaptive opponents and held-out evaluation; they do not establish performance against Pangram 4 or Slop Ninja's future incumbents [22,23].

The proposed contribution is a text-focused protocol connecting origin-detection competition to cleanup in a particular writer's voice, with independent preservation, readability, and requested-voice review. The transformation target combines removal of unwanted writing patterns with retention of the writer's individual habits and required evasion of Pangram and qualified subnet detectors; retired, authorized rewrites supply later detector-training examples, with source families kept separate from private evaluation. A artifacts are public; B miners retain their private transformation models. Both can sell hosted inference through an alpha-denominated market with explicit recipient encryption. This combination requires validation against the precedents above. The survey establishes neither uniqueness across all subnets nor superior performance by an implementation that has not yet been built.

3 Participants and the two kinds of work

A miners publish immutable, complete detector artifacts. B miners may retain private transformation models and register signed rewrite endpoints. Both register capabilities and authenticated protocol identities. A miners may separately offer paid detector hosting. Customers buy detection and transformation jobs through separately quoted offers. Validators execute the frozen A artifacts for authoritative benchmark scores, evaluate B outputs, and submit weights. Benchmark curators maintain permitted source material, production records, and hidden evaluation labels. A customer-facing gateway may relay ciphertext, but receives no decryption key.

Benchmark operator and coordinator denote administrative duties assigned to named curators and validators before each round: funding and managing the round, recording protocol-derived assignments, and obtaining baseline evidence. A pilot may combine these duties in one service. The task owner is the customer for private inference and the curator for a benchmark. Authorized evaluators and auditors are assigned validators; judges, readers, and authorized reviewers perform independent review for those validators. Automated judges require the qualification described in Section 8.3. The settlement adapter is the proposed contract software for customer escrow and alpha transfers, described in Section 7.

An *evidence preparer* is a hash-assigned validator who assembles the review dossier. A *certifying validator* belongs to the round's frozen eligible set and independently checks the evidence. In certificate rules, a *signer* is such a validator signing a particular final verdict. These duties may overlap; preparation alone grants no approval authority. Section 9 fixes the weighted quorum.

An A *model-credit entry* records performance credit for one detector's inference content after exact duplicates are coalesced under Section 4.1. It uses the canonical submitting UID and existing A payout policy; it creates no additional UID, token or payout pool.

Property	Customer inference	Subnet benchmark
Purpose	Serve a customer's request	Measure miners for emissions
Launch services	A: detection; B: transformation	A: detection; B: transformation
Source of text	Customer	Authorized benchmark collection
Default input recipients	Customer and assigned miner	Curator, assigned service miners, validators, commissioned reviewers
Evaluation	Customer acceptance	Validator execution of frozen A artifacts; independent B output review
Pangram submission	Excluded from the confidential job protocol	Required for every B benchmark
Payment	Customer's quoted fee	Performance-dependent emissions
Training reuse	No automatic reuse	Only under the collection's release terms

Publishing an A artifact does not grant its publisher access to private benchmark text. Validators execute it on authorized evaluation inputs; miners receive only the inputs required for their assigned tasks.

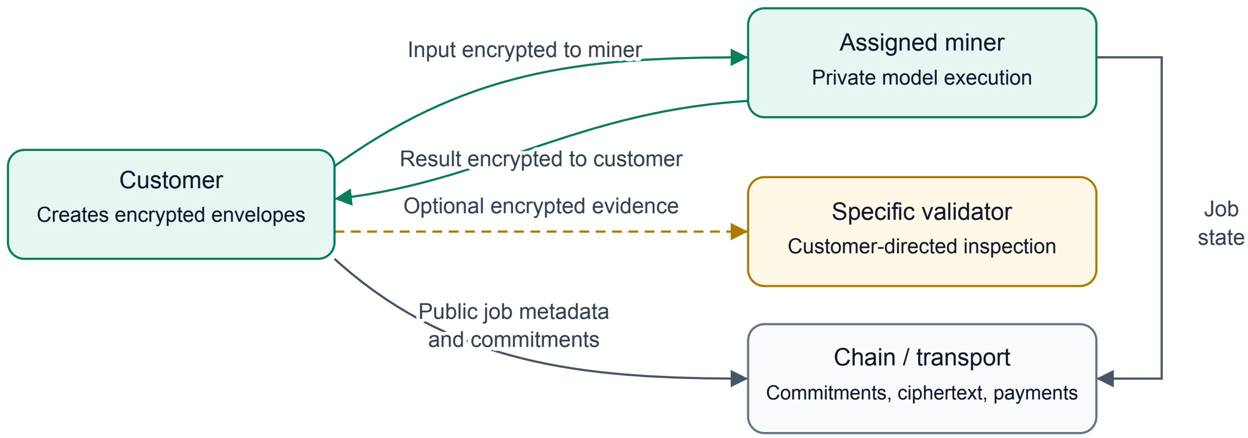

Figure 1: Customer inference and optional evidence disclosure. Only the addressed recipient can decrypt each envelope; endpoint handling remains a trust assumption.

Customer tasks do not become benchmark examples automatically. Validators must not obtain customer inputs through an audit exception, a shared support key, or an arbitration process. A customer may initiate a separate, explicitly addressed evidence disclosure as described in Section 6.3. This applies to both detection and transformation requests. An author's uploaded reference collection can be more identifying than the query itself and receives the same protection.

Miners may advertise specialties, prices, length limits, and delivery times. Mandatory training tickets assign an A requester to a B rewrite provider on authorized text. A contributes published artifacts; B runs them locally for training and submits its benchmark self-scores for independent validator replay. A operates no required scoring endpoint. The network evaluates observable service performance and published A computation. It cannot infer how many GPU hours a miner used, ownership of every component, or which private B weights produced a revision. Emissions create an incentive to improve training; they do not reimburse unverifiable training claims.

4 Model interfaces and starter architectures

The two launch interfaces, A for detection and B for transformation, define compatible jobs and outputs. Miners may change their internal architectures, grammatical representations, search methods, and training data without changing those interfaces. Sharing a foundation model between tasks is permitted. A detectors must be published as complete immutable artifacts; an update enters a later qualification round. B models and generation methods may remain private. A training recipes and data need not be public, subject to the publication rights required for the resulting artifact. Those terms must permit free access, local inference and reuse for training private B models; an artifact available only behind a paid inference API does not qualify. Optional A hosting may still charge for compute and delivery.

4.1 Author and origin detection

An author task supplies a query and reference works for a gallery of candidate authors. The response contains probabilities over the supplied author IDs. The starter compares full sparse word/grammar profiles; a learned text encoder with an author embedding head is the next architectural comparison. Reference prototypes support new customer authors without fitting a permanent classifier for each identity. Unknown-author rejection requires a separate calibrated output and withheld-author evaluation.

An origin task returns probabilities over documented production histories: human-only, model-only, and mixed human/model work. The production record includes revision order and assistance type. A model-to-model rewrite remains model-only under this initial label scheme. Pangram supplies a comparison, not the training label. The proposed encoder uses a separate origin head; author similarity must not be substituted for provenance.

ModernBERT-base [24], a 149M encoder, is a concrete starter candidate. Selection must compare it with the existing metric on new authors, topics, and registers. The subnet does not require that checkpoint or assume it will win.

Published artifact and execution policy. Each A artifact binds weights, features, tokenizer, preprocessing and postprocessing, calibration, reference pooling, executable dependencies, runtime, numerical precision, hardware requirements and output rules. Pin prompts and random generators where used. Qualification must establish a reproducible reference execution, including fixed tolerances and a resolution rule for numerical boundary cases. Validate and cache the qualified panel, its artifact hashes and thresholds before revealing the B task or allowing candidate generation. Validators execute that panel themselves for authoritative benchmark probabilities; a miner's reported score is not the scoring oracle.

The panel uses distinct UIDs and distinct inference-content hashes, excluding owner and signature metadata from the content identity. Identical content receives one model-credit entry and one possible panel seat; the earliest finalized accepted commitment wins, with canonical UID order for ties. Small changes and equivalent implementations remain a qualification problem. Select any reserve artifact before task issue. After issue, a runner outage permits execution of the same artifact on a reserved runner, not substitution of another detector or threshold. An unresolved reference execution voids the common comparison under the closure rules; it is neither B nonresponse nor an automatic pass.

Independent hidden author and production labels qualify each artifact. Include targeted-trigger controls: a published model can faithfully execute a rule designed to favor a colluding B. Publication verifies computation, not reliable detection or independent ownership. The isolated runner provides no network or undeclared state and excludes miner identities and assignment metadata from model inputs; text itself can still reveal task roles, so this does not prove blindness.

4.2 Author transformation and detector evasion

Inputs are the source, independent target-author samples, requested style changes, and a preservation brief. The brief identifies unwanted writing patterns to remove and intentional voice features to retain. Before generation, it fixes the target tonal intent using any requested style and supplied reference samples. A dramatic authorial-style shift can change tone while preserving the source's full meaning. Author matching covers lexical, grammatical and rhetorical habits in the requested register; generic polish alone does not satisfy it. Every B task pursues author matching and evasion against Pangram and the common subnet AI-origin panel defined in Section 1. The miner returns exact revised text or an abstention. Every B benchmark requires target-author references, style calibration and the complete detector evidence. Opponent selection and scoring are fixed by the benchmark, independently of the submitting miner. Author-directed revision of one's own or licensed writing and authorized detector-robustness research motivate this task. A detector pass does not replace the documented production history or any disclosure required for the intended use.

The proposed starter is an 8B decoder, initially Qwen3-8B [25], with a style adapter and a candidate-selection process. Named word and grammar deviations can condition the first version. A learned profile prefix is a later comparison, contingent on better results for unseen authors. Deterministic Rust edit rules can propose small changes; the decoder can propose broader restructuring.

Public A artifacts let B miners run local training queries without a protocol query quota, at their own compute cost. They can use published ensembles and retired A snapshots instead of depending on a peer's availability for every search step. Direct gradients are available only where the architecture permits them; differentiable feedback does not guarantee useful discrete text edits. Training tickets serve authorized rewrite requests from A to B under bounded limits. Benchmark submissions retain the frozen candidate and deadline rules. B can compute its own candidates' A scores before and after commitment; these public-model scores are not hidden feedback. The official comparison uses the one committed candidate and the frozen panel. Hidden labels, other miners' candidates and preservation-review evidence remain restricted.

B submits the complete self-scored probability vectors and declared scalar projections for every assigned A artifact, binding its hash, settings, exact source inputs and final candidate. Validators independently run those artifacts on the same inputs and compare the canonical outputs; they do not reproduce B's private generation process. A miners need not serve inference calls or help calculate these benchmark scores.

Train first on reviewed source/reference/brief/revision tuples. Then learn preferences among candidates that pass preservation review. Include originals, failed rewrites, copied reference passages, and fluent meaning-breaking edits as controls. Also include polished revisions that erase the writer's voice and detector-passing revisions that retain unwanted filler or formulaic phrasing. A miner's own style score may guide its search, but cannot approve its output for benchmark rewards.

Transformation briefs may specify audience, clarity, accessibility, organization, rhetorical force, and tone. A lower reading level or shorter text is useful only when the brief calls for it. Suitable text can remain unchanged. Validators use independent judges and reader audits to assess preservation, readability, and requested voice; reviewer ties and disagreements remain in the data.

Future comparisons of 4B, 8B, and 32B transformation models should use fidelity, reader preference, and cost per accepted revision. All checkpoint licenses and revisions must be pinned before a training release. The named starters are proposals, not trained subnet products.

5 Training data and evaluation separation

The two launch tasks need three linked data tables, with different supervision:

Table	Records	Supervision
Authorship	Multiple independent works per writer	Identity, date, genre, work family, provenance strength
Production	Original writing, generations, and revision chains	Recorded workflow, parent links, generator version, assistance order
Revision preferences	Sources, briefs, candidate edits, and reviews	Preservation, target voice, editorial preference, ties

Maintain distinct training, public development, private evaluation, and customer reference collections. Split authors before selecting vocabulary or fitting weights. Within evaluation authors, use independent earlier works as references and later works as queries. Keep every revision, translation, and excerpt of one source family in the same split. Hold out generator families and prompt families for origin evaluation. Match content, length, and genre across origins so that a detector cannot win by identifying the assignment format.

A artifact publication does not require releasing its training corpus or recipe. Qualification uses independently labeled hidden material, with targeted-trigger variants and controls for favoring a paired B. Models receive only declared text and reference inputs, not miner identities or assignment metadata. This removes an explicit cue without making the content blind to its production history or benchmark role.

B training can query published A ensembles and retired snapshots locally at the miner's compute cost. Measure transfer on fresh source and author families and later detector versions. Keep external Pangram observations and query counts, including failed attempts; success against an available A model alone does not establish transfer to Pangram or unseen detectors. Active evaluation material has no general training release; disclosing a source for B generation permits its declared task use. B can compute public A scores on that source and its own candidates. Hidden labels, unreleased sources and other miners' candidates remain restricted until their declared retirement and release.

Mandatory training tickets run in one direction: an assigned A requester supplies authorized source text and a brief, and its assigned B provider returns a constrained rewrite. Retain the source's production labels and keep source families within their declared splits. B obtains detector feedback by executing published A artifacts locally. Retired benchmark revisions can later supplement A training under their release terms.

Global Voices supplies a prospective pool of attributed modern prose under its default CC BY terms. Qualification must distinguish original writers, translators, shared accounts, and third-party material. PLOS articles can supply factual source material and citation constraints, while their coauthored nature makes them unsuitable as unqualified single-writer labels. Item-level rights and attribution remain part of acquisition. Global Voices policy [26], PLOS article policy [27].

Historical publication alone does not prove a human-only workflow. For stronger labels and useful editing supervision, the proposed pilot commissions 50 writers to provide twelve independent pieces each and two edits of supplied drafts. Common briefs across writers permit comparisons with similar subject matter; multiple subjects per writer test consistency beyond topic. Record collaboration and AI assistance, and obtain separate rights for training, private evaluation, and any redistribution.

If collection and review prove workable, expand to 500 writers: 300 training, 100 development, and 100 private evaluation authors. A further cohort is needed for unknown-author rejection and stronger confirmation. These are acquisition targets; the writers have not been commissioned and this paper commits no funds.

Generate model-only and mixed examples from permitted sources while retaining exact prompts, model revisions, outputs, and human editing records. Synthetic restoration pairs can start from a writer's original, create a generic draft, and train a return toward the original. Review whether that draft still contains the required information; otherwise restoration teaches invention. Keep such weak supervision separate from real writer edits.

Rewrites that fool Pangram or a subnet panel detector are hard examples for later detection rounds. Adversarial benchmark revisions retain their original production labels. Evasion does not turn model-origin text into a human-only training example. They may become later detector-training examples only after retirement from private evaluation and under agreed release terms. Preserve difficult human examples alongside them. Customer text, profiles, and outputs receive no automatic training or benchmark reuse.

The existing Blog Authorship Corpus experiments remain local noncommercial research and do not supply the commercial starter dataset. Private book and correspondence material is also excluded. Source manifests must distinguish text rights from loader-code licenses. The detailed acquisition plan [28] records the current source screening and outstanding qualification work.

6 Confidential customer inference

6.1 Assigned-miner encryption

For every customer job, the protocol gives input-decryption capability only to its assigned miner. The customer already possesses the input and may retain it. Validators, gateways, storage operators, other miners, and chain observers receive no input key or alternate recipient envelope. Models execute on the assigned miner's infrastructure; confidential jobs may not forward plaintext to an external model API, detector, or another miner.

Public A artifacts do not make customer inputs public or authorize validator replay of customer jobs. Validator execution for emissions uses the separately authorized benchmark collection. A customer may choose to run a public detector locally; a job sent through this protocol retains the assigned-miner input boundary. Public weights do not authorize sending customer text to Pangram or any other external service. Paid A hosting is optional and separate from artifact publication and protocol-assigned training requests. Its customer inputs retain the same confidentiality rules as private B jobs.

A miner advertises a separate X25519 encryption key in a hotkey-signed record binding the network, service, key ID, validity window, and job or offer scope. Signing and encryption keys have distinct roles. The customer verifies the miner identity and finalized key registration, reserves that miner's offer, and encrypts locally before sending any payload to the network. A gateway cannot select a substitute key after receiving plaintext.

The proposed HPKE suite uses X25519 with HKDF-SHA256 for key encapsulation, HKDF-SHA256 for key derivation, and ChaCha20Poly1305 for authenticated encryption. A customer signature authenticates the envelope and its job context; HPKE Base does not itself authenticate the sender. Bind the chain, subnet, contract, job ID, assignment generation, offer hash, recipient key ID, expiry, and ciphertext to the authenticated request. Check replay state and assignment validity before processing. These are application requirements around the HPKE primitive. RFC 9180 [29].

The encrypted payload includes the source, reference works, style profile, brief, private metadata, and a commitment opening. Public commitments use a versioned, unambiguous encoding and a fresh secret 32-byte salt. An illustrative commitment is

$$C = \mathrm{SHA256}\big(\mathrm{domain} \parallel \mathrm{context} \parallel \mathrm{salt} \parallel \mathrm{length} \parallel \mathrm{payload}\big). \tag{1}$$

Plain text hashes, filenames, report URLs, and descriptive task titles stay inside the envelope. Padding limits exact length leakage; public size buckets, timing, prices, and counterparties remain observable.

The result is encrypted to the customer's authenticated output key. The miner knows the result it generated, but supplies no result envelope to validators. The same rule covers probabilities, explanations, and change notes: detection results can reveal sensitive inputs too.

6.2 Assignment, recovery, and the privacy boundary

Each assignment binds one miner and one encryption-key version. Rotation cannot silently redirect an existing request. A retry with another miner requires the customer to authorize a new assignment and encrypt a new request. The protocol does not escrow recovery keys or re-encrypt inputs for a fallback miner. If the customer is unavailable, a failed job expires under its settlement terms. Reassignment cannot erase a previous miner's knowledge of text it already read.

Use short-lived, preferably job-scoped keys and an explicit retention policy. Ordinary recipient encryption does not establish that a miner deleted keys, plaintext, or logs. A compromised recipient key may expose recorded ciphertext. An assigned miner can copy plaintext after decrypting it; this design protects against unauthorized protocol recipients and observers, and assumes the chosen miner follows its data-handling terms. It does not prove secrecy against that authorized endpoint. Hardware attestation or computation over encrypted data would require a separately specified and evaluated design.

There is no automatic exception for customer support, quality disputes, subnet spot checks, or Pangram verification. A customer may separately choose to disclose material to another party. In particular, the subnet must not advertise a public Pangram report as compatible with exclusive miner access.

6.3 Customer-directed evidence disclosure

A customer can ask a specific validator to inspect a job by creating a new evidence packet encrypted to that validator's authenticated key. The customer signs a disclosure context binding the job, selected validator and key version, purpose, scope, and validity window. The packet can contain source passages, the delivered result, the miner's signed response, and relevant commitment openings. The customer selects what to send. Neither the miner nor a gateway can activate this access on the customer's behalf.

This grants the selected validator access to that packet, with no decryption capability for other jobs or undisclosed material. The validator must not forward it to other reviewers or providers without further customer direction. Expiry ends authorization for new processing; it cannot erase plaintext the recipient already obtained. Keep the packet, findings, and openings private; publish only any agreed decision or commitment needed by the protocol.

The validator checks which supplied evidence opens the original commitments. A redacted excerpt does not open a commitment to a complete document. Full openings or a separately specified selective-disclosure commitment scheme are needed for stronger binding. Incomplete evidence can support a limited opinion, but cannot establish fidelity of the whole job.

For detection, that validator may reproduce a published A artifact on the supplied material within the customer's disclosure scope. This grants no access to other customer text and does not verify execution of a private B model.

Inspection is advisory by default. A signed decision affects escrow only if the job's original terms gave that validator or selection policy authority and both parties accepted those terms. Choosing a reviewer does not itself change payment rights or extend deadlines. This route permits voluntary inspection while preserving the default assigned-miner input boundary.

7 Paid asynchronous jobs and on-chain coordination

The launch market offers A detection and B transformation jobs. A customers pay for compute and delivery while retaining the option to run the public model themselves. That alternative can constrain hosted A prices; the protocol prescribes neither a discount nor a price ratio between A and B. Both use the alpha-denominated market and the same customer privacy rules. B jobs pursue author matching and detector evasion through local generation and evaluation. A private job has no verified Pangram result unless the customer separately authorizes external disclosure; its objectives do not expand recipient access. Authorized benchmark rounds provide the routine external measurements under Section 11.1. The proposed job contract uses Subtensor's EVM to coordinate offers, escrow, commitments, and settlement. Inference prices are denominated in Slop Ninja's subnet alpha; actual model execution stays on miner hardware. Subtensor EVM guide [30].

7.1 Market pricing in alpha

Miners publish signed offers for bounded jobs. Each offer specifies a total alpha price, task and service version, input/output limits, delivery deadline, available capacity, quote expiry, and settlement terms. Customers select among compatible offers using price, benchmark performance, and delivery time. A higher-quality model or faster service may command a different price. There need not be one clearing price for every kind of inference.

Miners may revise their unreserved offers as demand, queue length, compute costs, and competition change. The subnet operator sets no base price, utilization target, or global price-adjustment coefficient. Miners choose their own pricing policies; customers decide whether the offers are worth buying. Alpha is the unit of account and proposed settlement asset. The inference market determines the number of alpha units charged independently of the alpha/TAO exchange market.

An atomic reservation consumes offered capacity and locks the quoted alpha amount and job terms. Later price changes apply only to later reservations. An expired or exhausted offer cannot be filled; repricing requires a new customer choice. The reservation therefore gives both parties a known price through execution and settlement, even while other offers change.

Quote discovery uses public service descriptions and the size/deadline information the customer chooses to reveal. Source text, reference writing, and the full brief remain encrypted for the selected miner. Obtaining competing offers does not distribute the plaintext to prospective bidders. If the selected miner cannot serve the bounded request, the job follows its rejection/refund terms; it cannot silently raise the price after decrypting the input.

7.2 Reservation and delivery

Subtensor's staking-v2 interface exposes allowances and stake transfers, and contracts can hold stake through their mapped coldkeys. The proposed escrow adapter would use those operations to custody unlocked, transferable alpha on the same subnet. An allowance alone does not fund a reservation. Alpha remains associated with a subnet and hotkey; settlement must account for that position and its native 10^{-9} precision. These primitives do not supply a finished job-escrow implementation [31,32].

An offer distinguishes the service price from network and transfer fees and names the fee payer. Funding succeeds only after the credited alpha principal satisfies the quote. Release and refund must reconcile actual balances, transfer permissions, and fees; incidental stake accrual must remain separate from job principal. The adapter and its accounting policy require validation before paid launch. Neither an alpha price feed nor an assumed ERC-20 interface substitutes for that work.

The proposed lifecycle is:

1. **Offer and reserve.** The customer selects a miner's signed offer, pins its key and job terms, and escrows the quoted alpha amount for a bounded input/output.
2. **Encrypt and submit.** The customer posts the salted input commitment and encrypted-delivery hash, then delivers the envelope to that miner.
3. **Accept.** The miner decrypts, checks the opening and limits, and accepts before the acceptance deadline. Failure to accept permits a refund.
4. **Execute and deliver.** The miner runs the private service, commits the result, and makes its customer-encrypted envelope available before the delivery deadline.
5. **Acknowledge or expire.** A customer acknowledgment releases payment. If there is no timely acceptance or delivery claim, the customer may claim a refund. A delivery claim without acknowledgment follows the bounded timeout policy below.

Bind acknowledgments and state transitions to chain, contract, job ID, role, nonce, and assignment generation. Bind the alpha asset's subnet ID and amount in native units as well. Authenticate the association between a Bittensor hotkey and its settlement identity. A hash or truncated address is not an authorization method. Enforce exactly one terminal settlement and explicit finality rules. A customer, miner, or keeper must submit timeout transactions; deadlines do not execute themselves.

Customers may deliver ciphertext through authenticated asynchronous endpoints and redundant encrypted storage. Small envelopes could instead travel through on-chain calldata or events, so miners can discover and complete jobs entirely through chain messaging. Payload-size, gas, and retention measurements must precede that choice. Storing ciphertext on-chain does not hide metadata or provide permanent secrecy after key compromise.

7.3 Payment without disclosing the job

A customer can consume a useful result and refuse acknowledgment. A miner can claim delivery of useless or undecryptable bytes. Public commitments alone do not distinguish those cases. Validators who cannot read the job also cannot adjudicate its meaning or writing quality.

For the first private-job pilot, use customer acknowledgment for payment and a fixed, bounded refund timeout for unacknowledged jobs. Publish that policy in the offer before acceptance. This assigns nonpayment risk to the miner, including when a customer has already consumed a valid result. Cap pilot job values and record the loss rate. It is an explicit service tradeoff, not trustless fair exchange. Public protocol evidence can resolve duplicate or late transactions; it cannot resolve a secret editorial disagreement.

Benchmark reputation can help customers choose a service without exposing their own work. It does not prove any particular unknown-input prediction correct. The customer-directed inspection route in Section 6.3 can support an agreed dispute policy. Its effect on payment must be fixed before job acceptance; a customer-selected opinion alone cannot redirect escrow. The initial pilot retains acknowledgment and bounded timeout refunds. Stronger private payment guarantees remain a launch research question.

7.4 Revenue and costs

Customer fees pay for the purchased service. Emissions reward performance on separate benchmark assignments. Customer volume does not directly increase emission rewards, because miners can transact with themselves. Paid successes cannot dilute failures on mandatory protocol service: the measured availability gate applies in aggregate and separately to each committed interface and assigned role's mandatory work. Its zero-credit and recovery-deficit rules are specified in Section 8.2. The zero-credit effect must be validated under the chain's payout lag; it cannot reclaim already distributed emissions. A fixed quote avoids charging for hidden, unverifiable internal token or search counts.

Miners pay their own training, inference, private search, and required candidate Pangram submissions for benchmarks. The benchmark operator or participating validators fund source-baseline reports from an explicit round budget; those reports can be reused across competing miners under the same task/version. Validators also bear A artifact distribution and benchmark execution, retrieval, hosting, and review costs. Subsidies, if used, must identify their funder and cap before assignments begin. This design assumes no free API-credit allocation or automatic on-chain reimbursement scheme.

Assigned B rewrite requests consume reserved provider capacity under the training-service obligation in Section 8.2; the recipient pays no per-call fee within that budget. Providers bear the compute cost as a condition of emission eligibility. A miners have requester obligations for those tickets, with no mandatory A inference endpoint. Authoritative A benchmark execution uses the validators' separately reserved budget. B miners may query public A artifacts locally without a protocol quota, bearing that compute cost themselves. This obligation covers neither Pangram's external charges nor unlimited hosted search. Bootstrap reference artifacts and execution need a separately named research funder. Confidential customer jobs cannot automatically forward inputs or outputs to Pangram or a peer miner.

8 Benchmark scoring and mechanism mapping

The initial subnet has two logical tasks and two chain mechanisms: A, author/origin detection, maps to mechanism 0; B, author-directed cleanup and transformation with detector evasion, maps to mechanism 1. Writing quality and meaning preservation are requirements within B, not a separate launch reward pool. Deployment must respect the documented joint UID/mechanism capacity constraint [33].

For an initial offline tournament, give A and B equal budget shares. Within A, balance author and origin tasks equally. Every B comparison jointly evaluates author-style matching and evasion of Pangram and the qualified subnet detector panel. These are pilot allocations, not an economically optimized split. Publish any live allocation and change policy before the scoring epoch. Isolated style or detector ablations remain diagnostics and earn no B emissions.

8.1 Detection

Validators obtain probabilities by executing the published, qualified A artifact frozen for the round. Its manifest binds the complete inference procedure and reference runtime; endpoint replies cannot replace these executions. Independently labeled hidden tasks determine author/origin performance. Publication establishes which computation is being scored, not the correctness of its predictions [34].

For a valid probability vector p and known one-hot label y, use the multiclass Brier utility:

$$B(p, y) = 1 - \frac{1}{2}\sum_{k}(p_k - y_k)^2. \tag{2}$$

Average query results within source groups and authors before the declared task mixture. Compare the aggregate with a fixed public baseline, then use positive improvement divided by that baseline's remaining headroom. If the baseline is perfect, the pool has no remaining improvement and is unallocated in the offline simulation. Clip after aggregate comparison, not separately for each lucky prediction. Raw Brier utility is unsuitable for direct payout normalization: a uniform prediction in a large author gallery can already have a high value.

Report retrieval accuracy, reciprocal rank, origin calibration, and human false-positive rates separately. Fix operating thresholds on development data. Unknown or weak production records cannot become hard labels merely because Pangram agrees with a miner.

Pangram remains the external quality anchor for A's origin task. Compare the evaluated A artifacts and Pangram on the same independently labeled hidden origin set under Section 11.1, and publish their performance gap and uncertainty. Continue this comparison after parity or superiority. Documented production histories determine reward labels; agreement with Pangram earns no separate utility. The fixed public probabilistic baseline above remains the reward normalization reference. Pangram's content fraction cannot directly replace that probability vector or supply author labels.

8.2 The adversarial feedback loop

Participation includes public A artifacts and a bounded training-service obligation. A miners request authorized training rewrites; assigned B miners provide revisions that follow the source and brief. The request is free at the point of use, with the provider bearing its cost in exchange for eligibility for task emissions. Expected emissions need not cover every miner's costs. The obligation concerns delivered service, not unverifiable training effort. Independent preservation, readability, and requested-tone review are required for B. A miner cannot approve its own revision for rewards.

Public A artifacts let B miners train and search locally. Those queries consume B's compute and have no protocol per-query quota. This local access does not create additional benchmark submissions or discharge an assigned service obligation. Validators separately reserve their own compute to grade the official comparison matrix. A earns quality credit from independent evaluation of its published artifact; B earns it from the committed revision. Section 4.1 defines canonical A execution.

Use released, authorized training sources and retain the production history, brief, and preservation status with each response. A later rewrite does not acquire a human-only label by defeating a detector. Unknown assistance histories remain weak labels; a private endpoint is not itself proof of model generation. The source-family splits, release rules and access boundaries in Section 5 govern this exchange. Private B weights can still be studied through served outputs; request quotas limit that exposure without preventing model extraction or retention by recipients.

8.2.1 Service budgets and deterministic pairing

Freeze a global budget, request schedule, interface qualifications, and capacity limits before each evaluation epoch. A artifact execution and B rewrite service have separate resource limits and capacity reservations. New UIDs and traffic do not increase the global budget, and serving volume earns no reward by itself. There are no counterparty nominations or voluntary pairing offers. Customer market quotes remain separate from this protocol-assigned work.

For each protocol-issued interaction, consume the next lifetime index i_u of the requester UID slot u. The counter is persistent for that chain and subnet, including through hotkey changes and UID reuse; each assignment also pins the current registration generation, hotkey, service version, and encryption key. A requester cannot skip indices, reset them, or issue empty calls merely to advance the counter. An expired assigned slot still consumes its index and counts as a requester failure if the requester abandons its obligation; retransmission keeps the same index, request, and peer assignment. New or replaced registrations must qualify before earning rewards; inheriting a slot counter does not inherit its former holder's performance record.

Let R_e commit the eligible, active service roster fixed for epoch e. For an A requester, rank each eligible B provider $v \neq u$ by

$$h_{u,i,v} = \mathrm{SHA256}(\mathrm{encode}(D, \mathrm{chain}, \mathrm{netuid}, u, i_u, v))\,, \tag{3}$$

where D is a versioned domain and encode is a canonical, length-delimited encoding. Break hash ties by UID. The first k distinct entries form the small permitted peer set. The first peer is assigned; later entries are prescribed fallbacks following recorded failure, not a menu from which the requester picks a favorable result. The pilot must publish k, capacities, and fallback limits before work begins. If the permitted set is exhausted, retain the failure; do not hash again. Hash-derived ranking and the fixed global schedule must respect provider capacity without allowing discretionary substitutions. Process capacity reservations in a canonical slot order; a request is issued only when its prescribed service windows fit the reserved capacity. Fallback stages have predeclared deadlines and minimum response windows, so later providers cannot inherit an impossible deadline. The ticket pins R_e for eligibility without mixing it into each peer's rank: changing one roster member does not reshuffle the relative order of all remaining peers.

A ticket binds the epoch, UID/index, roster root, prescribed peers and roles, permitted task, salted input commitment, size limits, deadline, and nonce. Anyone can check the scheduling metadata and sequential consumption against the finalized ledger. The text and commitment openings remain restricted to authorized recipients. Counterparty UID visibility does not establish the origin of the text. Validate authorization before inference; duplicates return the recorded response without another charge against the quota. Counter issuance and quota consumption require one authoritative, replay-protected ledger shared by validators, not independent per-validator counters.

This schedule is predictable. A miner can inspect future assignments, and another UID may have the same owner. Registration costs, bounded work schedules, and penalties for assigned failures constrain abuse; the hash does not prove operator independence or eliminate strategic abstention. Block height defines protocol-assigned slots and deadlines. Using a requester-chosen submission block as the peer seed would permit waiting for a favorable set. A short deadline narrows that opportunity without making a public block number unpredictable.

8.2.2 Availability from actual requests

The default Bittensor tempo is 360 blocks, approximately 72 minutes at 12 seconds per block; the actual subnet tempo and finalized epoch boundaries govern the schedule [35]. No heartbeat is required. Use authenticated outcomes of real assigned inference requests to measure availability.

The mandatory classes are A as requester of training rewrites and B as their provider. A artifact availability and performance are separate qualification requirements. The schedule must reserve real work for every activated mandatory class before that class can earn availability credit.

For miner u and measured epoch e, let $N_{u,e} = S_{u,e} + F_{u,e}$ count settled successes and attributable failures under Section 9. Retain unresolved obligations separately as $U_{u,e}$; they block eligibility without inventing a failure. If validator non-certification leaves a service obligation unresolved, even an honest miner loses earned emission eligibility for that measured epoch. If this affects every miner, no positive eligible skill remains and the pool follows the unallocated/burn policy below. An obligation is either submitting a scheduled request or answering a valid delivered request; the ticket fixes the responsible UID and role. The emission-eligibility gate is

$$L_{u,e} = \begin{cases} 1, & U_{u,e} = 0,\ N_{u,e} > 0,\ 10F_{u,e} \le N_{u,e}, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

More than 10% failures means zero earned emission credit in both A and B for that measured epoch. Exactly 10% passes this availability gate, while quality requirements still apply. With no assigned requests there is no availability evidence and no automatic pass. Report counts and failure rates by interface as well as in aggregate. Apply the same gate separately to mandatory protocol service for each committed interface and assigned role, and require every applicable gate to pass. Purchased traffic cannot dilute failures on mandatory work. Successful outgoing requests cannot dilute missed provider replies. A miner needs a positive assigned workload on each interface for which it seeks eligibility; apply a role-specific gate when the schedule assigns that role work. An answer after its deadline remains a recorded late answer. A successful fallback does not erase the preceding provider's decline or failure.

Only protocol-issued requests within the fixed input, output, capacity, and response-time limits enter N. Unsolicited traffic, duplicate retries, malformed inputs, and deadlines that cannot meet the declared service window do not count. Count a protocol-issued requester obligation even if the requester withholds its payload to avoid an unfavorable peer; do not count that undelivered payload as the provider's failure. Bind the request's measured epoch when it is issued; every counted deadline must fall within that epoch's closed measurement window. Requests due later belong to a later window and cannot be marked late early. Section 9 fixes complete ciphertext publication, validity certificates, fallback activation, and incident closure. A counterparty's accusation alone cannot establish a failure. Publish unresolved and void counts alongside settled work; neither retires recovery deficits.

Derive the next epoch's normal active roster from registration, qualification, and the completed prior window's request records. A small, deterministic probation allocation of real requests lets new or recovering UIDs establish availability without a self-declared online flag. Missing evidence can block normal admission without proving misconduct. Freeze each issued interaction's roster and ordered peers; mid-epoch declines or outages invoke its existing fallback list rather than recomputing the set. The probation budget and minimum workload coverage must be measured in the pilot and fixed before live use.

The zero-credit cutoff can remove the immediate incentive to finish an already failed epoch. For each mandatory interface and role, retain a recovery deficit alongside the epoch gate:

$$D_e = \max\{0, D_{e-1} + 10F_e - N_e\}, \qquad D_0 = 0. \tag{5}$$

Only authenticated, protocol-assigned work in that same class enters this update. Normal admission and emission eligibility require zero remaining deficit as well as the current availability and quality gates. Past surplus service cannot be banked; an epoch with no work cannot erase a deficit. Probation uses bounded real requests to permit recovery. Purchased jobs and other interfaces cannot retire the deficit, and key or service-version changes within a registration do not reset it. Dropping an advertised interface or role does not cancel its accrued deficit; every outstanding class must be cleared before normal admission. Replacement registrations follow their own qualification and probation; the design cannot establish whether their owners are different.

For 100 fixed obligations, 11 failures followed by 89 successful replies leave a deficit of 10. Abandoning all 100 leaves 900. Both cases earn no credit for that epoch, but require respectively 10 or 900 subsequent successful obligations to clear the deficit if no further failures occur. Capacity limits determine how many epochs recovery takes. This rule gives continued service a future eligibility benefit; it does not prove that every miner values future emissions above its costs.

The measurement window must close early enough for the relevant weight submission and commit-reveal deadlines. Publish which payout cycle settles each measured epoch. The proposed zero-credit rule cannot retroactively erase emissions already distributed by the chain. The adapter must demonstrate its effect on both mechanisms under actual weight-reveal delays and stale-weight behavior before launch; this paper assumes no custom slashing primitive.

8.2.3 Common evidence for transformation rewards

Per-miner peer sets allocate authorized B rewrite service for A training. Those training outputs do not replace the common committed benchmark or let a requester nominate scoring inputs. The benchmark scheduler reserves validator execution capacity for a common comparison batch using published A artifacts. This matrix consumes no miner service-ticket quota or lifetime interaction index.

For each shared comparison index, deterministically select three qualified A artifacts on distinct UIDs and a B batch from the frozen registry. Include the strongest qualified origin detector from the previous completed A round in the A panel; hash-rank eligible remaining A UIDs under a separate comparison domain to select the other two and the reserve order. Then hash-rank eligible B UIDs outside that panel to fill the published batch size. Filter the deterministic A reserve ranking against every UID in the B batch and the initial panel, then freeze the remaining reserve order and capacity. Distinct artifacts must have distinct inference-content hashes, with duplicate model-credit entries coalesced under Section 4. A reserve cannot later score its own B entry. Insufficient reserved capacity prevents issuing that comparison; it does not permit another B draw. Qualifications, sizes, capacity, tie-breaking, and the shared counter are frozen and reproducible. Miners cannot nominate a panel, join a preferred batch, or obtain another draw by declining work. Only exact UID exclusion is asserted; different UIDs may share ownership.

Rank the strongest A service by held-out origin Brier performance, subject to the declared calibration and human false-positive limits. Author-retrieval rank does not qualify an origin service. Every B entry faces the same three detectors, source texts, briefs, authorized target-author references, style artifact and calibration, Pangram policy, source-baseline evidence, and frozen settings. Every B benchmark requires these inputs. Validate and cache the entire selected A panel before revealing the B task or starting its generation window. Artifact reserves may be used only before that issue point. Execute a complete source and candidate matrix after B candidate commitments, using the same frozen artifacts and reference runtime. This is what matched assignments mean. The three-member pilot is provisional; its cost, coverage, and susceptibility to colluding UIDs require measurement before live rewards.

Apply the frozen-artifact and runner-recovery rules in Section 4.1. Unresolved execution follows the whole-comparison closure in Section 10. A B miner that fails to submit receives zero on its assigned work. A missing strongest-detector result prevents a claim of success against that detector. Deliberately triggered execution errors are a required pilot attack case.

The median panel improvement used below limits one arbitrary outlier when fewer than half the panel UIDs collude. It does not establish that bound or prove owner independence. Validators reproduce model behavior; the artifact qualification controls in Section 4.1 must test deliberate bias. One A result cannot alone determine B's utility, and A earns quality from independent labeled evaluation, not from how harshly it scores B. The strongest detector's result is also reported separately. Legitimate detector disagreement alone is neither fraud nor a replacement trigger.

Execute A under an isolated reference interface that excludes B UIDs, source/candidate role labels, scheduling metadata and undeclared state. Mix human and model controls. Text itself can still reveal its origin or purpose; interface restrictions do not prove blindness. Retain validator-signed execution records binding the model, runtime, inputs and outputs. Miner-selected local search records cannot replace this matrix. B submits self-scores with its final candidate, binding the assigned panel hashes, reference settings, full candidate probability vectors, required source baselines and scalar projections. These numbers record the miner's claimed result under the assigned configuration, so validators can identify and investigate discrepancies tied to that submission. They earn no separate credit and do not reduce required validator execution. Validators recompute the canonical outputs and compare the claims; verified results determine rewards, with mismatches retained. A claimed high score cannot skip another candidate's verification. Commit the complete evidence by the fixed block deadline; hash-assigned validators review every scored submission and all required checks at launch. Bootstrap rounds use published, explicitly funded bootstrap A artifacts until enough competitive models qualify, and are reported separately.

8.3 Preservation before revision rewards

Each editing task has a preservation brief and hidden review material covering claims, participants, quantities, negation, uncertainty, citations, attribution, and argument relationships. Review the entire revision. Positive utility requires preservation of the source's full meaning and conformity to the task's target tonal intent. That target may differ from source tone when the requested authorial style calls for a change. Validators also check readability and tone under the fixed rubric in Section 10. Before issue, the brief identifies required cleanup of unwanted filler, repetition or formulaic phrasing under R, and intentional voice features to preserve under T, including any specified informality or idiosyncrasies. Cleanup must retain the protected information. These are B acceptance constraints. A certified failure sets $G = 0$; certificates for all required passes set $G = 1$. Every task requires tone review, including a request that supplies no separate tone instruction. An unrequested tone reversal can defeat the writing brief while retaining literal claims, so we require this vote and measure its added review cost and void exposure in the pilot.

Readers compare anonymized revisions in hash-derived order without detector or miner scores. Maintain unchanged sources and qualified editor baselines. Automated judges join reward allocation only after agreement and failure modes have been measured against separate reader judgments. Instructions embedded in a candidate are evaluated text, not evaluator instructions.

Unresolved judgments at the fixed close void the matched comparison for the whole B batch under Section 10. Miner nonresponse remains a zero on assigned work. A validator outage can still erase comparison credit or block epoch eligibility when a service certificate remains unresolved. Section 10 describes the residual risk that a miner induces semantic disagreement to void its batch. Publish coverage and missingness so that conditional accuracy cannot conceal excluded difficult cases.

8.4 Transformation utility

Let $V \in [0, 1]$ be the exact normalized positive improvement in target-style fit defined in Section 10. A frozen public evaluator returns integer source and candidate scores; private miner predictions cannot change them. Validate the evaluator against independent reader comparisons before live use. Topic copying, reference copying, and meaning-breaking edits must fail that validation. No miner-defined embedding can be its own reward oracle.

For Pangram evasion, let $F_r(t)$ be the reported AI-plus-assisted fraction in required observation r for text t. The initial benchmark requires three distinct completed reports for each source x and candidate z. Improvement using the least favorable pair among those submitted observations is:

$$\Delta_P(x, z) = \min_r F_r(x) - \max_r F_r(z). \tag{6}$$

The strict target additionally requires $\max_r F_r(z) < 0.10$. Three reads of one report count as one observation. The 10% cutoff is the project's declared engineering target. Any tolerance used to compare repeated reports must be reported separately and cannot widen that success criterion. Use the declared provider version and repeat policy for both sides; report sensitivity to provider variation.

Miners may pay for additional search and report attempts before commitment. Scoring uses the committed reports, including variation in their provider results. The protocol does not require disclosure of every private attempt or a fresh validator-paid Pangram inference call. A reported pass applies only to the committed observations, not to every future measurement of that text. Selection can bias the submitted set when provider results vary. We accept this limitation to keep external Pangram costs with the submitting miner. Validators independently execute the frozen A panel against the committed candidate. Our probe found equal document fractions across fresh requests for three tested texts, with auxiliary-score variation, but no observations within two percentage points of 10%. It establishes neither general determinism nor threshold stability [36].

For panel detector j, let $H_j(t) = p_j(\text{model-only} \mid t) + p_j(\text{mixed} \mid t)$, using its validator-verified probability vector under the frozen policy. These are document-origin probabilities; Pangram's F_r is a content fraction. Define $\Delta_j(x, z) = H_j(x) - H_j(z)$ and $\Delta_S(x, z) = \text{median}_{j=1}^{3} \Delta_j(x, z)$. Retain each detector's raw responses and improvements. The pilot evasion utility is

$$E(x,z)=\begin{cases}(\Delta_P(x,z)+\Delta_S(x,z))/2, & \Delta_P(x,z)\geq 0 \text{ and } \Delta_S(x,z)\geq 0,\\ 0, & \text{otherwise.}\end{cases} \tag{7}$$

The strict detector pass additionally requires $G=1$, $\max_r F_r(z)<0.10$, a majority of panel detectors satisfying $H_j(z)<\tau_j$, and a pass against the strongest qualified prior-round origin detector at its own threshold. Calibrate each τ_j on independent development data at the declared human false-positive operating point and freeze it before the round. These thresholds are not automatically 0.10. Report strongest-A and panel pass rates separately; the strongest-A result is not a sole veto on incremental utility. Neither Pangram improvement nor panel improvement compensates for regression on the other aggregate axis.

When the required evidence is resolved, the sole B utility is

$$U_B(x,z)=\begin{cases}G(V+E)/2, & q_z\geq q_x,\ \Delta_P\geq 0,\ \Delta_S\geq 0,\\ 0, & \text{otherwise.}\end{cases} \tag{8}$$

Check these raw conditions before awarding credit: clipping a negative improvement to zero in V or E cannot conceal a regression and allow the other component to earn a reward. Both author fit and detector evasion are mandatory parts of every comparison. The semantic gate requires preservation of full meaning, readability and target tonal intent; neither component can compensate for a failed field. Unresolved evidence follows the common closure rule. A declared unscorable candidate earns zero B utility under Section 10.

Incremental utility can reward improvement in one component while the others remain unchanged. It does not certify an absolute author-fit threshold or the strict detector target. Report author fit and detector passes separately from incremental utility; a strict pass must also meet the raw nonregression conditions above. An already suitable source can stay unchanged without receiving an invented improvement bonus.

Validate this proposed utility against blinded reader comparisons before live reward use. It must order useful revisions above omissions, unsupported strengthening, irrelevant padding, and needless edits.

8.5 Aggregation and failure policy

Fix assignments, task mixtures, baselines, and deadlines before submissions. Balance related examples within source groups; extra cheap submissions do not increase a miner's share. Apply the availability and zero-deficit gates to each miner's earned A and B skill for the measured epoch. Then normalize eligible positive skill within each task pool and combine using the published shares. The chain mechanism split carries the corresponding aggregate task allocation.

If no miner beats a baseline, record that pool as unallocated in the offline simulation. The proposed live fallback directs that pool's weight to a verified owner-associated hotkey whose miner incentive is withheld by the native burn/recycle rule [37]. Pin its UID, registration generation, owner association, and the burn setting before weight submission; validate them again for the payout cycle. The live configuration must admit a single nonzero weight and a full-weight destination. The latter weight cap is root-controlled; owner control alone cannot establish this precondition [38]. Failure to meet these conditions blocks launch of this adapter. Do not substitute zero weights or leave a previous earned vector active.

Withheld owner-directed incentive can reduce the subnet's later emission share. Burning is therefore an economic cost, and a zero-incentive epoch has separate native redistribution behavior [39]. Quantization, registration changes, native reveal lag and the fallback must be demonstrated against the pinned runtime. A frozen application certificate specifies the intended weights; native consensus still determines the resulting payouts.

Execution and incentive limits. The two native task pools above reward independently evaluated A artifact performance and B output utility. Section 13 records earlier private-panel and shared-settlement experiments and why they do not establish conserved native payouts for this design. Public A models can still favor colluding B outputs, and public weights make copying possible. Qualification, duplicate handling, execution budgets and actual native settlement must be tested before live rewards. The design does not claim to eliminate all profitable self-dealing.

9 Mandatory-service evidence and closure

The proposed `service-evidence-v1` mailbox covers bounded, protocol-assigned rewrite requests from A requesters to B providers. A models are public; B generators remain private. The requester counters, provider assignments and availability gates are defined in Section 8.2. The one-ticket exercise below specifies transport and review of a B rewrite. It implements neither an artifact runner nor the complete common source/candidate matrix. An emission-bearing benchmark needs a separate frozen execution schedule and capacity budget. Private customer jobs retain their paid lifecycle and never enter mandatory-service counts. Inputs remain encrypted only to the selected miner; a customer may separately disclose evidence to a chosen validator. Full encrypted payloads are published on chain; a hash or HTTP locator alone is insufficient. Publication establishes bytes, sender authentication, and timing. Private validity and quality require the certificates below. This mailbox is a specified pilot, not an implemented throughput result.

Authoritative A execution follows Section 4.1; the common benchmark matrix and its separately funded capacity follow Section 8.2.3. The mailbox certificates below establish transport and declared service validity. They do not replace benchmark execution or semantic review.

Frozen authority. Before revealing task contents, exclude the exact assigned A/B UIDs from the round's eligible validator set. Bind every remaining UID, registration generation, hotkey, signing and encryption key versions, and integer native effective consensus stake, including native alpha/TAO scaling, to one finalized state snapshot. Its manifest names the block hash, runtime and storage schema, weight field and units, complete records, and total weight W. Initial loading and later snapshots require verified finalized state proofs or a chain-verified native snapshot. A relayer's assertion alone is insufficient [40].

Every final validity, quality, or incident verdict requires signatures whose distinct frozen weights sum to w, with $3w > 2W$. Assume dishonest weight is strictly below $W/3$ in this conditional set. Missing keys, nonresponse, recusal, and later key changes never shrink W. Three hash-assigned preparers organize evidence; they cannot authorize a verdict. Honest validators independently check evidence and sign at most one verdict per stage. Conflicting strict quorums would share more than $W/3$ weight; a detected conflict halts affected settlement. Signatures bind the verdict itself, evidence, snapshot, stage, and scope.

Fixed pilot. The instance manifest supplies actual chain/contract addresses, verified snapshots, authorizations, service/schema hashes, and funded reservations. Missing bindings, zero eligible weight, or insufficient qualified key and verification capacity prevent issue. At native epoch start H, issue at most one authorized rewrite ticket from an A requester to a B provider in a 360-block measurement epoch, with three prescribed B providers total. Freeze these settings before issue:

Parameter	Value
Input / output cap	4,096 / 8,192 bytes
Ciphertext chunk / publication cap	4 KiB / 8 MiB per stage
Global epoch publication cap	64 MiB, including all witness copies
Request publication / certificate	$H + 12$ / $H + 24$
B provider response window	36 blocks
Certificate window per attempt	12 blocks
Preparer commitment / private opening	$H + 180$ / $H + 192$
Final verdict and incident close	$H + 204$

Attempt $j \in \{1, 2, 3\}$ starts at $H + 24 + 48(j - 1)$, publishes by start plus 36 blocks, and certifies by start plus 48 blocks. The final B attempt cutoff is $H + 168$. Early declines do not advance later windows. Reserve all possible bytes, gas, response and verification capacity; insufficiency prevents issue and does not authorize another draw. One ticket provides a bounded transport pilot, not availability evidence for every UID. Settlement names a later native payout cycle and its actual commit/reveal configuration; no claim is made that reveal delays fit within the unused part of this epoch.

DATA, then witness. Use HPKE base mode with X25519/HKDF-SHA256/ChaCha20Poly1305, fresh single-shot ephemeral material per recipient, and separate registered sender signatures [29]. Deterministic CBOR records reject duplicate keys, indefinite encodings, tags, and floats [41]. A salted payload commitment and encryption context bind protocol, chain, contract, ticket, stage, attempt, both UID generations, and recipient key version. Input DATA copies address the requester and three prescribed providers; response copies address the requester and actual provider. Curator authorization covers those recipients and the complete validator set. Hidden labels and review answers remain in validator-only evidence packets.

First fix and publish the signed DATA manifest and all ciphertext chunks. Then form a witness containing that DATA digest, plaintext, salt, and each DATA envelope's ephemeral generation material. Encrypt the witness separately to every frozen validator and publish those complete ciphertexts under a second manifest referencing DATA. Validators reconstruct each DATA encryption and compare the exact bytes and recipient bindings, as well as authorization, commitment, and schema. Decrypting only their own copy would not check the miner's copy. DATA never references a witness hash; witnesses do not contain their own encryption secrets. This avoids a hash cycle. Generation witnesses remain confidential and need implementation/security review. Their certificates do not assert that every validator decrypted its witness copy. Public ciphertext remains exposed to future key compromise; authorized archives retain complete replay evidence.

Transitions and attribution. Issue permanently consumes the requester index and activates its obligation. One immutable manifest fixes each stage; identical retries are idempotent and conflicting signed manifests remain evidence. A requester's decline, absent complete publication, or certified invalid input is its failure. An authenticated decline included by the stage's publication cutoff is terminal for that stage. Retain later bytes as evidence, but reject an acceptance certificate that ignores the decline; late declines cannot reopen closed accounting. Only a timely global input-acceptance certificate activates the first provider at its fixed start. No input acceptance means no provider obligation. A provider's decline, absent publication, or globally certified invalid response permits the next fixed fallback; successful fallback never erases preceding failures. Unresolved validity stops dependent work without another draw. Late answers remain late.

Complete publication without a global validity verdict closes unresolved; recipient accusation alone cannot assign fault. Keys must be qualified before the snapshot. Key loss afterward cannot blame a sender whose encryption was certified correct; rotations and UID reuse cannot change issued tickets. Certified delivery of a B response does not establish its revision quality. A request-publication failure remains in its requester ledger and cannot alter a benchmark artifact or computed result. Validator runner failures never enter this rewrite-service ledger. Missing service-validity certificates do enter it as unresolved obligations, which block epoch eligibility even without miner fault (Section 8.2). B rewards evaluate the submitted revision without requiring replay of the private generator. Insufficient final review evidence voids the shared quality comparison consistently, without deleting attributable service failures. The detailed record and certificate rules are specified in the companion service-evidence document.

Counts and finality. For each applicable A-requester or B-provider class, retain activated obligations $A = S + F + U + V$: success, attributable failure, unresolved, and platform void. Unused reservations are separate. Let $N = S + F$; require $U = 0$, $N > 0$, and $10F \leq N$ for the class and aggregate availability gates. Publish all counts. Keep $D_e = \max(0, D_{e-1} + 10F - N)$; neither unresolved nor void work retires a deficit. Require zero outstanding deficits and current qualification; customer jobs and other classes cannot dilute these gates. A artifact publication and qualification are separate eligibility gates.

Only canonical finalized inclusion counts, with deadline equality timely. Subtensor separates block production from GRANDPA finality [42]. A local timeout or censorship allegation does not prove a platform exception. Normal operation assumes fair inclusion and sufficiently prompt finality. A finality stall delays observing closure without extending block-height deadlines. A strict global incident certificate must be included by $H + 204$, even if its block finalizes later, and identify the affected interval and every intersecting ticket/comparison. The rule voids all activated obligations in that scope, including successes and failures, and cannot selectively pardon a miner, shrink W, or reopen closed epochs. Without that certificate, normal inclusion rules apply; missing verdicts remain unresolved. Gas, history retrieval, proof-verified snapshots, and the declared native settlement timing must be demonstrated before this EVM pilot issues work [30].

10 Style measurement and final review

This section defines the style utility V and semantic gate G for B transformation. Every B task requires preservation of the source's full meaning, fulfillment of the task's target tonal intent, and readability. These constraints create no separate writing-improvement service or reward pool. Positive utility requires independent semantic certification; style proximity or detector evasion cannot compensate for a preservation failure. Numerical measurement uses a public frozen artifact. Semantic judgments are explicit oracle inputs certified under the validator-weight assumption in Section 9.

Validators obtain authoritative benchmark origin probabilities by executing the published A panel under the artifact-freeze and execution rules in Sections 4.1 and 8.2.3. They compare B's submitted vectors with canonical outputs bound to the same artifact, runtime and input. B models remain private; rewards assess their committed output text. B miners can compute public A scores during search, while hidden labels and unreleased semantic-review evidence remain restricted.

Frozen style measurement. Style measurement is required for every B benchmark, alongside Pangram and the frozen subnet detector panel. Each task supplies authorized target-author references and a frozen style calibration before issue. Before task issue, publish a permitted word/grammar author-distance artifact and pin its hash, parser and feature schemas, model parameters, reference-pooling rule, executable/runtime, numerical operations, rounding, resource bounds, and test vectors. Its distance function $d_\theta(t, R)$ returns a nonnegative integer for text t and target-author references R. The instance manifest must supply these concrete bindings; this specification does not select or claim to have trained the artifact. Private B generator weights do not define this public style evaluator.

Freeze a nonempty calibration multiset $C = (c_1, \ldots, c_M)$ of integer distances from development queries to their attributed authors' independent reference works. Keep development source families separate from fitting and evaluation, record provenance and hashes, and use the same artifact and reference-pooling rule. The task pins its calibration ID before candidate generation. Retain duplicate distances. With $Q = 1{,}000{,}000$, define the integer style fit

$$q(t, R) = \left\lfloor Q \, \frac{2\#\{m : c_m > d_\theta(t, R)\} + \#\{m : c_m = d_\theta(t, R)\}}{2M} \right\rfloor. \tag{9}$$

This empirical rank gives exact ties half credit. Smaller distance cannot reduce q. It is a calibrated distance rank, not an authorship probability or a certificate of preserved meaning.

For source x and candidate z, evaluate both against the same R, artifact, and calibration. Let $q_x = q(x, R)$, $q_z = q(z, R)$, and define

$$V(x, z) = \begin{cases} 0, & q_x = Q, \\ \dfrac{\max(0, q_z - q_x)}{Q - q_x}, & q_x < Q. \end{cases} \tag{10}$$

Equal fit earns zero improvement; a saturated source has no remaining headroom. Preserve the exact rational numerator and denominator through aggregation. Every B comparison requires $q_z \geq q_x$, even though V clips negative improvement. Task mixtures and final credit rounding remain part of the frozen allocation policy.

Score the source and references successfully before issuing a B task. A candidate that receives the declared deterministic `unscorable_candidate` status earns zero B utility, since style nonregression cannot be verified. Execution disagreement or unavailable shared evidence cannot silently supply a score: it remains unresolved under the common comparison-closure rule. A numerical failure alone does not establish attributable service nonresponse. Validate the frozen artifact against copying, topic substitution, and meaning-breaking controls before funds depend on it.

Miners can also optimize directly against the public style artifact. Before live rewards, test adaptive searches that maximize V under Section 14. Among revisions that pass P, R, T, higher V must still track independent reader judgments of author fit; the semantic gates alone do not validate that ranking.

Three semantic fields. The source's full meaning must remain intact. Before generation, resolve the target tonal intent from the brief and, where supplied, the requested authorial style and authorized reference samples; freeze it in the task rubric. A dramatic change of authorial style can authorize a different tone without a separate tone instruction. Use source tone as the default for aspects the request leaves unchanged. Review the whole candidate for meaning, including information not enumerated in the checklist. Tone review applies the frozen rubric and its source-tone defaults throughout the candidate; reviewers cannot add new stylistic preferences after generation. Every certifying validator records one verdict, `PASS`, `FAIL`, or `ABSTAIN`, for each field:

- P, preservation: retain the source's full meaning, including all claims, entities, quantities, negation, uncertainty, attribution, citations, and argument relations. Changes to expression must preserve that information. Unsupported material additions, omissions, altered meaning, or prohibited reference copying fail. Greater stylistic force cannot strengthen factual certainty or obligations.
- R, readability: introduce no material readability regression against the source and satisfy the brief's stated readability requirements, including required cleanup of identified unwanted filler, repetition or formulaic phrasing. Shorter text or simpler vocabulary alone supplies no pass.
- T, tone: fulfill the frozen target tonal intent and every tone and audience constraint. Preserve the intended voice features, including informality or idiosyncrasies, specified by that target. This field always requires a semantic vote, including when no separate tone instruction appears in the request.

Insufficient evidence or unresolved interpretation, including uncertainty about the target tonal intent, requires abstention. Failure ballots identify rubric items and bound source/candidate spans where applicable. Preference ties and extra editorial praise earn no additional utility. No detector probability substitutes for these semantic fields.

Authority and timing. Three hash-assigned preparers assemble the dossier; their vote count cannot finalize a judgment. Every final signer receives authorized benchmark evidence, checks the relevant source, references, brief, candidate and rubric, and attests the semantic field itself. The review client presents semantic evidence first; honest signers commit their field ballots before inspecting detector scores or numerical style results. Full-set authorization is not a cryptographic blinding guarantee. Checking three preparers' signatures or tally is insufficient. Reuse the verified frozen validator snapshot and integer effective weights from Section 9, including its conditional assumption of strictly less than $W/3$ dishonest eligible weight. Positive W, qualified recipient keys and complete capacity reservations are required before issue.

For the pilot epoch starting at H, preparers commit by $H + 180$ and open privately by $H + 192$. Evidence corrections close at $H + 196$; they may correct references to already committed evidence, but cannot change the candidate, brief, evaluator, or calibration. Each validator then commits its one final ballot per field by $H + 198$, opens it privately by $H + 202$, and the global field certificate must be included by $H + 204$. Deadline equality is timely; finalized inclusion, incident scope and payout timing follow Section 9. Reserve all witness publication and verification work before issue. Under the pilot response schedule, candidate certification occurs by $H + 72$ through $H + 168$, leaving 30 to 126 blocks before the ballot commitment cutoff: approximately 6 to 25 minutes at 12 seconds per block. Certifiers must review the authorized source and candidate as evidence becomes available; waiting for the preparers' private opening would leave only six blocks. These cutoffs do not establish measured review throughput.

The first canonical timely commitment fixes that validator's ballot for the field. Identical retries are idempotent; a conflicting signed ballot is retained as evidence and cannot replace it. Late, absent, or invalid openings contribute no agreeing weight. Keep missing, abstaining, recused, and nonresponsive validators in the frozen denominator W. A `PASS` or `FAIL` certificate requires distinct eligible signers of that verdict with total weight w satisfying $3w > 2W$. Honest signers never authorize conflicting final verdicts; conflicting strict certificates halt affected settlement.

Deterministic closure. At the close, set $G = 1$ only if all three fields have a valid `PASS` certificate. Set $G = 0$ if any field has a valid `FAIL` certificate. Otherwise the judgment is unresolved. Conflicting certificates take precedence and halt settlement. Missing candidate submissions remain attributable miner nonresponse under the service-evidence rules.

An unresolved candidate judgment voids that matched source/brief comparison for the entire B batch. It cannot selectively remove a difficult candidate or erase independently attributable service failures. Retain all ballots and disagreements; do not draw replacement judges to obtain a preferred verdict. This rule makes outcomes reproducible from certified inputs. It does not require honest readers to interpret every passage identically or turn quorum agreement into semantic proof.

Whole-batch closure prevents validators from using unresolved judgments to exclude only one competitor's candidate, at the cost of giving B submitters a way to disrupt the comparison. A chosen candidate can split honest judgments or induce abstention so that neither PASS nor FAIL exceeds $2W/3$ on a required field. If no other field receives a FAIL certificate, that candidate voids the comparison. Dissent alone need not cause a void: enough agreeing failure weight certifies $G = 0$. A dishonest minority below $W/3$ also cannot prevent a certificate when all remaining weight agrees and participates. An unresolved semantic judgment adds no service failure or recovery deficit; the submitter still pays its generation and any report costs. A UID expecting zero skill could therefore remove rivals' positive skill from one comparison and increase its owner's normalized share through sibling UIDs scoring elsewhere. This economic risk remains unresolved. The bounded attack study in Section 14 must measure its frequency, collateral loss and owner-level payoff before live rewards.

The versioned adjudication record binds the task and comparison IDs, source/candidate/reference/brief commitments, evaluator and calibration hashes, exact q_x, q_z, V, rubric fields, evidence root, validator snapshot and W, ballot commitments/openings, field certificates, deadlines, G, and any unresolved or void reason. Records use the canonical encoding and authenticated encrypted evidence paths of Section 9. Evidence must reach every authorized certifier within the funded publication limits. Customer jobs never enter this process automatically. A execution records additionally bind the complete published model manifest and canonical runtime configuration. Certificates attest scoped judgments and faithful computation. They do not prove semantic correctness or require reproduction of private B generation. A artifact loading, numerical replay, and full-matrix capacity require their own implementation checks beyond the one-ticket A-requester/B-provider rewrite mailbox pilot.

11 Detector evidence and assigned audits

Pangram's API provides asynchronous tasks, analyzed text, document fractions, version information, and optional public dashboard links. The returned text may differ through preprocessing, so validators must bind evidence to the analyzed bytes rather than a preview. The local reader currently requires exact equality; a normalization policy would need its own version and review. Pangram API [43].

Our synthetic probe obtained a publicly retrievable report containing the analyzed text and scores. An independent caller retrieved it without a new paid inference. The observed web endpoint is undocumented and needs an availability, retention, and usage agreement before production dependence. We have not established a provider signature, immutable model identity, or one unique canonical result per text. Recorded probe [44].

For every B benchmark, a miner commits its final candidate and required report IDs before the evidence deadline. Every validator in the round's frozen certifying set receives the source, candidate, report URLs, score projections, and commitment openings in a separate encrypted evidence envelope. A signer must retrieve the provider's record and inspect the exact text, fractions, version, time window, and required preservation evidence; preparer signatures cannot substitute for that inspection. Report IDs and URLs remain encrypted in chain payloads and absent from public plaintext logs. Anyone who obtains a public URL may still retrieve its content; encrypting the locator protects our delivery path, not Pangram's hosting. Curator authorization must cover all frozen certifiers. Private customer jobs remain outside this benchmark; a customer's optional disclosure to one chosen validator does not authorize wider benchmark disclosure.

Pangram states that it does not train Pangram 4 on customer API data. Private B weights prevent direct downloading, but do not prevent independent black-box study through purchased outputs or leaked examples. The subnet must not assert otherwise. Provider verification and any reuse of provider labels also need suitable service terms. Pangram model card [10].

Public A artifacts can be inspected by Pangram and other providers. Pangram remains an independent external origin check when a subnet detector is unavailable, but its result cannot stand in for author attribution, preservation or a missing A result in the joint benchmark. Report external and subnet-detector outcomes separately; missing A evidence cannot become an adversarial pass. Customer text is not sent to Pangram without the customer's separate authorization.

For public A scores, use the complete comparison matrix, B self-score bindings and signed execution records in Section 8.2.3. Every final signer independently verifies the required computations under Section 4.1; a coordinator's signature alone is insufficient. Section 9 specifies complete on-chain ciphertext publication, separate confidential generation witnesses, and fixed validity/closure certificates. A report URL, response hash, or recipient accusation alone cannot satisfy that service-evidence contract. Its runtime implementation still needs validation independently of the report reader.

11.1 Pangram cadence and detector parity

At launch, schedule one joint B benchmark round in each scoring epoch. Freeze its admitted miner cohort, comparison coverage, report budget and deadlines before issue. Every credited B candidate requires both the author-fit evaluation and the complete Pangram/subnet evidence defined in Section 8.4. The submitting miner funds its three candidate reports; the benchmark operator funds the three-report shared source baseline and separate parity studies. Reserve full validator inspection and execution capacity before issue. The 24-task pilot must test feasibility; it has not established that this workload fits the declared epoch.

This cadence governs scored benchmark submissions. Miners can train and search locally between rounds using public A artifacts. Private customer text retains its exclusive-recipient policy. A prior report, another candidate's score or an untested estimate cannot fill a missing Pangram observation. Apply the existing evidence and closure rules when required reports are unavailable, and apply current service-availability and recovery gates at settlement.

Pangram also remains A's permanent external origin-detection quality anchor. In each A scoring epoch, compare all admitted A artifacts with Pangram on the same fresh, independently labeled hidden origin set. The operator funds the shared Pangram observations and reserves their budget before issue; this cost does not multiply with the number of A artifacts. Freeze the reference set, provider version, repeat policy, operating points and analysis before evaluation. Publish performance gaps between A and Pangram, uncertainty and coverage, including missing provider evidence. Missing evidence cannot support a current parity finding. Reuse a B report only when its exact text and observation satisfy this independent evaluation protocol; a set of miner-selected successful evasions cannot replace the labeled reference set. Matching or exceeding Pangram does not retire this continuing A evaluation.

Reducing B's per-candidate Pangram checks requires demonstrated detection parity. Before opening the evaluation set, freeze the actual A champion and panel policy proposed for substitution. Compare them with Pangram on the same independently documented human-only, model-only and mixed production histories, including fresh adversarial B revisions and held-out writers, source families and registers. Keep those production labels after successful evasion. Measure detection sensitivity at declared human false-positive operating points and require the frozen human false-positive limits to hold, including predeclared difficult slices. Fix noninferiority margins, sample sizes, report aggregation and uncertainty calculations before collecting results; confirm the finding on successive unreleased evaluation sets. Agreement with Pangram alone is insufficient.

Pangram's content fraction and A's document-origin probability have different meanings. Any calibration comparison must first map them to the same labeled event using separate development data. Bind parity findings to the tested versions, policy, populations and dates; changes in provider version or observed deterioration require renewed evaluation. Publish aggregate results and certify the finding under the frozen validator quorum.

This launch version retains Pangram in every scored B round, including after a parity finding. Parity permits consideration of a future policy with less frequent external checks on B candidates. That policy must retain periodic B checks and A's continuing Pangram quality comparison, and specify its interval, scoring of unchecked candidates and response to drift before activation. It may not alter an issued comparison or claim a Pangram pass for text the provider did not check. There is no automatic switch based on miner claims of parity.

11.2 Assignment and audit sequence

A training requesters and B providers are paired by the lifetime interaction counter and frozen eligible roster. Benchmark batches, common detector panels, and preparer assignments use a separate, shared comparison counter and distinct hash domains. These schedules are publicly predictable. At launch, review every scored submission and all required checks, including whole-revision preservation. There is no hidden audit sample and no application-level external beacon dependency.

For each scored submission, rank eligible validator UIDs by the hash of the versioned review domain, chain, subnet, shared comparison index, submitting UID, and candidate preparer UID. Use canonical encoding, UID tie-breaking, three distinct evidence preparers, and an ordered reserve list. Exclude the exact miner UIDs involved in that comparison before freezing its certifying validator set and native effective-consensus-weight snapshot. The coordinator cannot choose preparers or obtain another draw; different UIDs can still have the same owner. Reserve preparation and full-set verification capacity before issue.

Preparers organize checks and retain independent commitments/openings. Final certificates follow the frozen authority and strict weighted quorum in Section 9. They attest the actual verdict; preparers have no separate approval authority.

1. Freeze the eligible epoch roster from qualification and actual request outcomes, service capacities, global task budgets, index-allocation rules, deadlines, and authorized certifying validator roster. Verify its native weight and identity records against the finalized snapshot specified in Section 9. Commit the question pool and source Pangram baselines. Publish fixed block deadlines for submissions, complete evidence, preparer commitments, private openings, dispute resolution, and weight submission.
2. Issue A-requester/B-provider rewrite tickets in the protocol schedule. Consume consecutive lifetime indices, derive their ordered peers from the pinned roster, and record responses, declines, expiries, and fixed fallbacks. Training feedback uses released, authorized material; active benchmark labels and detailed review material remain private.
3. Use the shared comparison index to determine the common A panel and B batch under the frozen policy. Exclude every B UID from A reserves before freezing pre-issue artifact reserves. Validate and cache the complete selected A artifacts, reference runtimes and thresholds before revealing the B task or starting generation. Accept one final B candidate and selected Pangram-report commitment per assigned task. The predictable opponent schedule does not allow a second draw or a revised submission after commitment.
4. Execute the required common panel matrix in the restricted reference runner, retaining control outcomes and execution failures. Apply Section 4.1 for runner recovery and Section 10 for unresolved computations. Record any missing strongest-detector comparison. Finalize all evidence commitments before the fixed evidence deadline; a favorable score is never grounds to switch provider or discard an observation.
5. Publish complete recipient-bound DATA ciphertexts, followed by separate confidential witness/evidence envelopes to every frozen certifier, under Section 9. Retain finalized inclusion proofs and enforce key versions, expiry, replay state, and publication deadlines. Preparers commit answers before peer openings, then open privately to the frozen certifiers. Every signing validator inspects the evidence, verifies the DATA-generation witnesses and report/panel bindings, and signs the exact verdict under the strict weighted threshold. Close unresolved records under the fixed rule before the declared settlement cycle; missing evidence never grants approval. Withhold hidden labels, other miners' candidates and detailed review material until retirement. B can calculate its own public A scores throughout local training and search.

Retain every review and disagreement. A missing preparer follows its fixed reserve policy; a missing global certificate closes unresolved under Section 9. Neither defaults to approval. Full coverage prevents a miner from concentrating defects in a publicly predictable unreviewed subset. It does not establish validator honesty or guarantee that readers detect every error. Weighted certificates require the stated honesty bound, and stake identities alone do not establish it. Reduce task volume to fit the funded preparation, ciphertext publication, and full-validator inspection budget instead of reducing coverage or the certificate denominator. Any later sampling policy needs a separate security argument and measured cost benefit.

Subtensor's timelocked commit-reveal protects validator weights from immediate copying. Use its supported SDK/runtime path, including any beacon dependency within that native mechanism; this proposal adds no separate beacon wait for peer allocation or auditing. Weight plaintext eventually becomes public; customer inputs must not be encrypted with that eventual-reveal mechanism. Use the recipient encryption described above and do not hard-code a CRv4 schedule. Bittensor commit-reveal [45].

12 Trust assumptions and adversarial behavior

The chain can enforce authenticated accounting and commitment consistency. Recipient encryption limits who can decrypt protocol deliveries. Counter and roster commitments permit reproducible assignment of peers, comparison batches, and reviewers. Every scored submission receives the required review at launch. Each guarantee has a narrower scope than useful, confidential writing inference.

Risk	Proposed response and remaining assumption
Customer text reaches another miner or gateway	Encrypt locally to the pinned assignment; no alternate keys or automatic reassignment
Assigned miner leaks decrypted text	Minimize retention and enforce service terms; endpoint nondisclosure is not cryptographically proved
B miner retains private weights but fabricates quality claims	Score submitted revisions against the brief and independent review; internal generation remains unproved
B supplies false self-scores	Validators execute every assigned A artifact on the bound source and final text and compare canonical vectors and scalar projections; B's report does not authorize its own reward
Miner copies reference content or removes difficult claims	Full preservation review, copying checks, and explicit exploit controls
B miner chooses a candidate that prevents a semantic quorum	Whole-batch voiding limits selective exclusion by validators but can erase rivals' skill; measure induced voids and common-owner payoff before live rewards
Miner's private search and retry costs are unverifiable	Miners fund their own attempts; score committed reports under fixed deadlines; do not reimburse claimed private costs
Validators collude or score inconsistently	Every final verdict needs strictly more than two-thirds of frozen eligible weight. Assume less than one-third is dishonest; every signer inspects the evidence and missing votes remain in the denominator
Curator or coordinator leaks labels or selects a biased task set	Independent provenance review, overlap checks, and source-group splits; commitments constrain later changes to the frozen pool and baseline evidence. Label secrecy and honest curation remain trust assumptions
Published A detector favors a transformation miner	Hidden qualification, targeted-trigger tests and a common frozen panel executed by validators; replay verifies computation but not the absence of colluding model behavior. Distinct UIDs do not prove independent models or owners
A model is copied under additional identities	Pin complete artifacts and give identical inference content one model-credit entry and panel seat; small changes or a new hash do not establish an independent contribution or owner
Pangram changes or serves conflicting records	Pin the reported interface/version, retain observations, and suspend affected comparisons under a fixed policy
Customer consumes work and withholds payment	Bounded pilot exposure and explicit miner nonpayment risk; private fair exchange remains unresolved

Risk	Proposed response and remaining assumption
Miner manufactures paid demand	Keep customer revenue separate from emission scoring

The quorum assumption applies after excluding the exact task participants; it must hold in that conditional validator set. It prevents conflicting certificates when honest validators sign once. It does not prove their semantic judgments correct or a published detector accurate. Authoritative A outputs come from validator execution of the frozen artifact, with evidence checked under the declared runtime and output rules. The quorum still assumes enough honest validators perform those checks. Native Yuma also processes each validator's submitted weights independently. A valid benchmark certificate cannot force a dishonest validator to submit its prescribed weight row. Public replay does not change the native payout transformations studied in Section 13; model behavior and validator weights still need separate incentive analysis.

The benchmark also needs private source rotation, duplicate detection, and delayed release of retired examples. A published model may recognize a repeated test or contain a trigger shared with B. B can inspect public A weights and optimize against them locally; that access does not validate the resulting text or its provenance. Changing the artifact, benchmark or judge after observing a candidate would invalidate the declared comparison. Publish protocol versions and aggregate outcomes while keeping customer material, hidden labels, unreleased sources, other miners' candidates and private review evidence restricted. B can calculate its own public-A scores; those probabilities cannot be treated as concealed feedback.

Training tickets bound aggregate B rewrite work requested by A. Public A models can be queried locally at B's own compute cost. The predictable hash schedule does not prevent common owners from obtaining multiple UIDs or exploiting occasional favorable assignments. It provides verifiable allocation without an ownership registry. Service failure rates need authenticated request and delivery evidence; the strict greater-than-10% rule cannot be based on a recipient's complaint alone. Predictable assignments remain mandatory, and requester abandonment consumes both an index and its failure allowance. The permitted 10% allows selective refusal. Timely valid rewrites may still fail quality evaluation; availability does not certify quality. The 90% gate and recovery deficits apply to issued A-requester and B-provider obligations. Separate role counts prevent cheap successes from diluting failures. A persistent recovery deficit makes abandoned work delay later eligibility even after the current epoch's credit is lost. It cannot make an operator who plans to leave value future participation. Cache and validate the frozen A artifacts before task issue. A runner outage can use reserved capacity for the same artifact; an unresolved reference execution requires the declared common-comparison closure and cannot select a different detector after seeing B's candidate. A has no required inference endpoint. Training-service qualification uses actual A-requester and B-provider obligations and deterministic probation work, without heartbeats. Optional paid hosting has its own customer terms.

13 Existing evidence and implementation status

The current Rust representation uses word occurrences and grammatical features from spaCy annotations. It scores complete sparse distributions and learns selected coordinate corrections. It does not require an LLM to enumerate or measure deterministic edit candidates. The current author/detector models and editing rules support experiments for A detection and B transformation; a demonstrated end-to-end product remains to be built.

The table reports top-1 accuracy averaged within each author, then equally across authors, then across three seeds. Query counts describe unique inputs scored in every seed; these author-macro percentages are not pooled fractions of those counts. Evaluation protocols [46,47].

Experiment	Author-macro top-1	Scope
Blog author retrieval	56.30%	600 evaluation authors in six 100-author galleries; 1,206 queries
Global Voices transfer	73.81% combined; 77.98% lexical-only	Fourteen provisionally qualified authors; 58 queries; topic/editorial confounds
Compact document projection	32.82% Blog; 25.00% Global Voices	Development-selected linear encoder; previously evaluated test corpora
Remove unselected contributions from old scorer	40.83% Blog; 29.17% Global Voices	Conditional diagnostic retaining old pooling and calibration

The Global Voices comparison removes grammar terms from the frozen Blog-trained scorer without refitting or renormalizing any remaining term. That removal improves accuracy by 4.17 percentage points on this sample, reversing the grammar contribution's benefit on Blog data. The experiment does not identify why: it has no matched-topic control and does not isolate register, editorial practice or extraction effects. This supports testing feature weights across registers; it does not establish that grammar is generally unhelpful [47].

The projection experiment shows why 3,557 learned coordinate weights must not be mistaken for a sufficient fixed input vocabulary. The full model also retains unselected word and grammar contributions. Neither author retrieval nor these feature ablations establish meaning-preserving editing or detector evasion. In a separate edit diagnostic, replacing *may* with *must* improved author proximity despite changing the claim. Independent preservation checks are therefore central to the design. Author projection study [46], edit counterexamples [48].

Implemented components include Rust tools for corpus collection and evaluation, portable author scoring, offline challenge and submission checks, a public Pangram report reader, and an HPKE prototype that encrypts receipts for assigned validators. The receipt prototype relies on caller-supplied trusted assignments; it is not a customer-request protocol or deployed key registry. Receipt implementation [49].

Design history of the stress tests. The following experiments include earlier designs with private A inference endpoints, reciprocal A/B service obligations and three-reviewer majority judgments. Some fixtures therefore use two tickets and four service roles. The current design publishes A artifacts, requires training service from A requesters to B providers, and certifies judgments through the global weighted quorum (Sections 4, 8.2 and 9). The complementary-credit and shared-pool policies tested below remain experimental settlement proposals. Their arithmetic and counterexamples constrain the design without becoming adopted payout rules.

13.1 Assignment and accounting stress tests

A reproducible Rust experiment uses SHA-256 rankings for 100 synthetic miner UIDs and a separate, already-eligible 100-UID reviewer roster. It evaluates 10,000 scheduled slots for each of three fixed seeds, with fixed cartel shares from 0% to 50%. The table gives pooled counts for the 20% case. Peer rates concern cartel requesters; panel and reviewer rates concern all modeled comparisons [50].

Event	Count / opportunities	Rate
Cartel peer, mandatory first assignment	1,128 / 6,000	18.80%
Cartel peer, requester chooses among three	2,875 / 6,000	47.92%
Cartel A majority, strongest A honest	1,179 / 30,000	3.93%
Cartel A majority, strongest A colluding	10,250 / 30,000	34.17%
Cartel majority among three reviewers	3,060 / 30,000	10.20%

Counting selective requester skips against each interface's 10% limit raised favorable partners among delivered requests to 20.89%, while leaving their absolute count unchanged. Aggregate-only accounting let 34 of 60 cartel UID-epochs pass despite exceeding an interface's limit. These results support mandatory first assignment, persistent consumed indices, and separate service gates. The recovery calculation in Equation 5 also passes the experiment's boundary checks; its incentive effect remains conditional on valuing future participation.

Colluders can answer every request on time and reserve dishonest answers for known favorable committees. Under the modeled majority rule, complete review coverage then fails to establish correctness. Exact without-replacement calculations lower the reviewer-majority frequency from 10.11% for three reviewers to 1.47% for nine at the same fixed 20% share, with three times the review calls. Larger committees reduce this modeled frequency without proving an honest majority.

These are assignment and accounting tests, not measured model accuracy, profitable attacks, or an equilibrium analysis. The simulation assumes fixed membership and reserved capacity; it does not model adaptive UID acquisition, shared-detector fallback, delivery disputes, private-model execution, or native payouts. It also does not test the public A artifact runner required by the current design.

13.2 Settlement counterexamples and certificate boundaries

A second isolated Rust experiment checks 1,921,920 binary-probability cases and 693 three-class cases for a proposed complementary integer allocation. For fixed admission and a pot c, assigning B $\lfloor cb \rfloor$ and A its remainder conserves c in every admitted case. Rejected joint admissions award neither endpoint. This verifies accounting in the synthetic model; it does not verify the quality oracle or imply alpha-payout conservation [51].

With separate normalization, moving an allied pair's raw credits from $(12{,}000, 0)$ to $(9{,}000, 3{,}000)$ increases its combined allocation from 10.00000% to 10.83591%. Even fixed complete weight rows leave a counterexample: with 70% honest and 30% malicious validator stake, a simplified native clipping and rank-normalization model increases the pair's miner-incentive share from 13.15789% to 13.56362%. Its combined policy credits remain 12,000. The simulation omits native fixed-point details, registration filtering, dividends, pruning, and future emission changes. It disproves an inference from conserved credits to conserved payouts; it is not a measured deployed exploit.

The abstract certificate model also checks 1,332 admissible pairs of weighted vote sets without a conflicting strict quorum at or below one-third dishonest weight when honest validators never double-sign. Exactly one-third can block liveness by withholding. Above one-third, an explicit roster admits conflicting certificates through double-signing. This vote-set model does not implement the mailbox's first-commitment ledger. Removing missing voters from the denominator breaks the safety argument even below that boundary. Service fixtures cover unresolved obligations, requester fault before provider activation, replay accounting, the 10% boundary, and persistent recovery deficits. Seven focused tests, formatting, and Clippy pass for this isolated experiment. These checks supply neither a production ledger nor an economic equilibrium proof.

13.3 Shared-pool and dividend follow-up

A third isolated Rust experiment considers settling complementary A/B credits in one native pool. With identical complete post-filter honest rows, fixed honest stake s, fixed outside credits and coalition credit sum c, the maximum coalition miner incentive is $c/[s + (1 - s)c]$. It depends on the sum rather than the internal A/B split. The experiment checks 3,245,130 malicious-row cases and explicit maximizers. This invariant concerns the optimized miner-incentive envelope; it does not establish total economic receipts [52].

Historical validator dividends give a counterexample with 20% coalition stake and distinct miner and validator UIDs. Moving honest A/B credits from $(.05, .15)$ to $(.15, .05)$, while outside credit stays $.8$, preserves combined miner incentive but increases the dividend of a coalition validator with an asymmetric bond history. The replay uses unchanged production math function bodies from pinned Subtensor revision `67dcf7f`, explicit sparse-branch orchestration, stored bond quantization and integer epoch allocation. In a synthetic budget of one billion units, coalition miner-plus-validator allocation rises from 126,786,990 to 137,501,008 under legacy bonds. A separate Yuma3 fixture with Liquid Alpha disabled rises from 125,824,352 to 131,439,278. These are function-level replays with supplied histories, not SDK runtime executions or observed operator profits. The coalition is assumed to economically own the modeled validator receipts.

Using current, fully clipped legacy bonds removes the displayed history attack. The ideal maximum combined share is $(c + 1 - s)/2$ when $c \leq s$, and $c(2 - s)/[2(s + (1 - s)c)]$ otherwise. A finite grid and explicit maximizers support this calculation. The native swap fixtures give identical units under that configuration, but a 19-split sweep retains rounding sensitivity: the matching-row strategy spans 4,884 units out of one billion. This is not a bound over all native rows. The proposed settings remain a candidate, with native filtering, registration changes, integer encoding and future incentives still to establish.

Separate bounded-cost pot fixtures test irreversible funding and per-ticket default bonds. They deter the enumerated independently attributable defaults under the declared cost bound; an unobservable private execution shortcut remains outside that result. Nine focused tests, formatting and Clippy pass. These findings constrain settlement claims and motivate the selected public A policy. They do not test its artifact runner, copying incentives or detector qualification.

The project does not yet demonstrate a reliable sub-10% rewriting model, calibrated production-origin service, reviewed commercial training release, authenticated live benchmark ledger, customer escrow contract, or private serving endpoint. The results justify a measured pilot and constrain its design; they do not establish readiness for emissions or customer funds.

14 Development and launch criteria

First, run a local tournament with 24 rights-cleared tasks across the two launch interfaces, A detection and B transformation. Evaluate detection on known-label author and origin tasks against fixed probabilistic baselines. For transformation tasks, compare unchanged text, deterministic edits, and one general editor under the same briefs, target-author references and Pangram/subnet detector panel. Measure both objectives on every B benchmark; isolated objective ablations are diagnostics and earn no B emission credit. Three hash-assigned preparers assemble dossiers, including intentional omissions, uncertainty changes, and padding. Frozen validators independently assess B's preservation, readability and tone; strict weighted certificates finalize those fields. Preference remains a diagnostic. Freeze the public style artifact and calibration before comparison and publish aggregate agreement, coverage, cost, and failure cases. Reserve and measure the per-epoch joint B benchmark and continuing A quality-reference schedule, including operator-funded Pangram observations, in Section 11.1. Test that no B credit can omit author references, style calibration or required detector evidence, and that raw regression on either objective cannot be hidden by a clipped score. A parity study must use independent production records and the frozen comparison protocol before any proposal to reduce B's Pangram call frequency. A's external comparison continues after parity and after any such B policy change.

Evaluate cleanup and author consistency through blinded comparisons with held-out writing in the same register and, where available, human revisions under the same brief. Record removal of unwanted patterns, retention of intentional voice features, and preservation failures separately from detector scores. Before claiming indistinguishability, predeclare the comparison task, acceptable difference margin and sample size, and report uncertainty intervals against that margin. Failure to find a significant difference in a small sample is insufficient.

The 24-task pilot intentionally prioritizes protocol and failure measurement. Reserving four B comparisons for the attacks below leaves thin coverage of ordinary transformation performance. Report those results descriptively. General performance and indistinguishability claims require a larger, separately powered study across writers and registers.

Before live reward use, include adaptive style-metric attacks in that larger study. Give the search procedure the published evaluator and allow repeated edits selected to maximize V under a declared budget. Compare with unchanged sources and ordinary editing, with matched generation/compute budgets for competing revision methods, on held-out source families and writers. Retain query histories and all candidates; report metric gains, blinded author-fit judgments, semantic failures and search costs, including disagreement among revisions that pass P, R, T. Freeze the evaluator and rubric for each comparison; any correction enters a later validation round. This study is separate from the 24-task protocol pilot.

Within that pilot, reserve four B source/brief comparisons for chosen-candidate semantic attacks: ambiguous attribution or negation, ambiguous preservation of qualifications, borderline readability, and disputed compliance with tone constraints. Each comparison uses three committed candidate variants: an intended clear pass, a clear violation, and an attempted unresolved judgment. Include a requested dramatic tonal shift as an intended pass, and certainty changes, unrequested tone reversal and loss of required informality among the violations. Briefs without separate tone instructions still require target-tone certification, using the requested authorial style or the source-tone default as appropriate. Full meaning preservation remains mandatory in every case. Freeze the rubric, validator weights, reviewer assignment and candidate budget before generation. Review all variants independently under the same field rules; retain their ballots even when common closure voids the comparison. This gives twelve candidate evaluations within the fixed pilot budget, separate from chosen-input execution-failure controls.

Record field-level `PASS`, `FAIL` and abstaining weight, unresolved and whole-batch void rates, review time, candidate/report costs, and positive competitor skill excluded by each void. Replay the frozen allocation with and without each attack entry to measure changes in a simulated common owner's share when sibling UIDs score in other comparisons. Compare with benign ambiguity and infrastructure-only non-certification under the same budget. Publish those counterfactuals as diagnostics, alongside the actual whole-batch outcome. This bounded study cannot establish resistance to adaptive attacks. Live-reward acceptance must address any profitable induced voids; naming the risk does not supply a defense.

In parallel, use synthetic jobs and test funds to verify assigned-miner encryption, customer-only result delivery, key substitution rejection, replay prevention, expiry, acknowledgment, and exactly one settlement. Exercise miner outage, wrong keys, unacknowledged consumed results, and attempted automatic reassignment. Neither a validator nor a gateway should have a decryption path. Measure encrypted payload sizes and chain costs before enabling on-chain transport. No real customer text is needed for that pilot.

Use the writer pilot to establish acquisition and review costs. Refit public starter models only on data cleared for the intended release. Evaluate author and origin models on new sources, then train the transformation adapter after enough reviewed pairs exist. Reward scaling requires demonstrated judge reliability and preservation performance, with uncertainty reported on unseen sources.

The one-ticket rewrite mailbox exercise in Section 9 is a transport and attribution pilot. It does not supply the complete common three-A validator-execution matrix or qualify every UID for emissions. A full tournament must reserve its expanded traffic and global review workload under a separately frozen schedule before issue. The 24-task offline tournament is not a demonstrated 360-block throughput claim.

Before a live subnet launch, resolve the supported Pangram retrieval/billing integration, key and assignment registry, replay ledger, public A artifact qualification and reference execution, protocol-derived counterpart and reserve assignments, service-ticket authorization and consumption, measured B rewrite-service quotas, delivery-dispute attribution, fixed measurement, evidence, review, and weight deadlines, and complete review coverage for every scored submission, empty-pool weight policy, and mechanism allocations. Exercise exhausted quotas, duplicate tickets, concealed common control, strategic refusal, and same-artifact runner recovery. Verify the cached panel and thresholds freeze before B task disclosure; no post-issue detector substitution is permitted. Verify that execution failure cannot become B nonresponse, an automatic pass or a private request's reassignment. Validate the greater-than-10% failure gate in aggregate and on each mandatory service interface and assigned role, with exactly 10% passing and zero workload supplying no automatic eligibility. Test requester abandonment, registration changes, deterministic probation, and minimum fallback response windows. Demonstrate the zero-credit rule and persistent recovery deficits, including same-class probation and resistance to resets through key or service-version changes. Verify settlement under native weight-reveal delays and payout cycles; already distributed emissions cannot be clawed back by this proposal. Before paid customer launch, separately establish acceptable nonpayment losses, retention terms, encrypted delivery reliability, and alpha escrow behavior under finality and timeouts. Verify quote expiry, capacity reservations, exact alpha accounting, release and refund permissions, and fee handling. These requirements must be concrete protocol parameters or measured acceptance criteria before funds depend on them. Reproduce the frozen style scores, field certificates and common void decisions from archived inputs. Demonstrate proof-verified validator snapshots, complete mailbox and witness publication, fault attribution and finality handling. Verify B self-scores through independent validator execution. Demonstrate complete artifact availability, canonical content identity, reference-runtime agreement, independently labeled qualification and targeted-trigger controls. Test exact-copy coalescing, near-copy incentives, chosen-input execution failures and full-validator replay cost. B's local public-detector feedback must be available during the pilot; measure transfer to fresh source families, later A versions and Pangram.

The next deliverable is a small reproducible tournament and private-job pilot, followed by revised parameters based on observed failures and reader judgments. The whitepaper remains a design draft until those results support deployment.

References

[1] It's AI. Text detection subnet. https://github.com/It-s-AI/llm-detection. Accessed September 9, 2026.

[2] It's AI. Mining and model FAQ. https://github.com/It-s-AI/llm-detection/blob/main/docs/FAQ.md. Accessed September 9, 2026.

[3] BitMind. GAS: Generative Adversarial Subnet. https://github.com/BitMind-AI/bitmind-subnet. Accessed September 9, 2026.

[4] BitMind. GAS generative mining guide. https://github.com/BitMind-AI/bitmind-subnet/blob/main/docs/Generative-Mining.md. Accessed September 9, 2026.

[5] Macrocosmos. Finetuning Subnet: Incentive overview. https://github.com/macrocosm-os/finetuning. Accessed September 9, 2026.

[6] Chutes. The Chutes starter guide. https://chutes.ai/docs/guides/starter-guide. Accessed September 9, 2026.

[7] Chutes. Private AI inference: What it means and how Chutes makes it verifiable. https://chutes.ai/news/private-ai-inference-verifiable-confidential-llm-serving. Accessed September 9, 2026.

[8] Targon. Infrastructure. https://targon.com/infrastructure. Accessed September 9, 2026.

[9] Targon. Confidential compute technical paper. https://targon.com/releases/intel-whitepaper. Accessed September 9, 2026.

[10] Pangram. Pangram 4 model card. https://www.pangram.com/research/model-card/pangram-4. Accessed September 9, 2026.

[11] GPTZero. Technology and API frequently asked questions. https://gptzero.me/technology. Accessed September 9, 2026.

[12] Grammarly. Introducing voice features. https://support.grammarly.com/hc/en-us/articles/23153676821773-Introducing-voice-features. Accessed September 9, 2026.

[13] Wordtune. Wordtune Editor explained. https://support.wordtune.com/en/articles/8995735-wordtune-editor-explained. Accessed September 9, 2026.

[14] QuillBot. Paraphraser modes. https://help.quillbot.com/hc/en-us/articles/35854318883351-What-are-modes-in-the-Quillbot-Paraphraser-and-how-do-I-use-them. Accessed September 9, 2026.

[15] Undetectable AI. Humanize documents. https://help.undetectable.ai/en/article/humanize-documents-1nehls2/. Accessed September 9, 2026.

[16] Anthropic. How Claude's text watermark works. https://www.anthropic.com/news/claude-text-watermark, 2026. Published August 14; updated September 1; accessed September 10, 2026.

[17] Anthropic. How Claude marks AI-generated content. https://support.claude.com/en/articles/16266773-how-claude-marks-ai-generated-content. Accessed September 10, 2026.

[18] Google DeepMind. Watermarking AI-generated text and video with SynthID. https://deepmind.google/blog/watermarking-ai-generated-text-and-video-with-synthid/, 2024. Published May 14; accessed September 10, 2026.

[19] Substack. How can I detect AI on Substack? https://support.substack.com/hc/en-us/articles/50891130623508-How-can-I-detect-AI-on-Substack. Updated August 9, 2026; accessed September 10, 2026.

[20] Katy Gero, Chris Kedzie, Jonathan Reeve, and Lydia Chilton. Low Level Linguistic Controls for Style Transfer and Content Preservation. In *Proceedings of the 12th International Conference on Natural Language Generation*, pages 208–218. Association for Computational Linguistics, 2019. https://aclanthology.org/W19-8628/.

[21] Jillian Fisher, Skyler Hallinan, Ximing Lu, Mitchell L. Gordon, Zaid Harchaoui, and Yejin Choi. StyleRemix: Interpretable Authorship Obfuscation via Distillation and Perturbation of Style Elements. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 4172–4206. Association for Computational Linguistics, 2024. https://aclanthology.org/2024.emnlp-main.241/.

[22] PAN and ELOQUENT. Voight-Kampff Generative AI Authorship Verification 2024. https://pan.webis.de/clef24/pan24-web/generated-content-analysis.html. Accessed September 9, 2026.

[23] Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. In *Advances in Neural Information Processing Systems*, 2023. https://arxiv.org/abs/2303.13408.

[24] Answer.AI and LightOn. ModernBERT-base model card. https://huggingface.co/answerdotai/ModernBERT-base. Accessed September 9, 2026.

[25] Qwen Team. Qwen3-8B model card. https://huggingface.co/Qwen/Qwen3-8B. Accessed September 9, 2026.

[26] Global Voices. Global Voices attribution policy. https://globalvoices.org/about/global-voices-attribution-policy/. Accessed September 9, 2026.

[27] PLOS. Licenses and copyright. https://journals.plos.org/plosone/s/licenses-and-copyright. Accessed September 9, 2026.

[28] Slop Ninja. Miner training sources and acquisition plan. https://github.com/sam0x17/slopninja/blob/77b32b3/docs/miner-training-sources.md. Repository revision 77b32b3.

[29] Internet Engineering Task Force. RFC 9180: Hybrid Public Key Encryption. https://www.rfc-editor.org/rfc/rfc9180.html. Accessed September 9, 2026.

[30] Bittensor. Subtensor EVM guide. https://www.bittensor.com/docs/guides/evm. Accessed September 9, 2026.

[31] Bittensor. Staking V2 precompile. https://www.bittensor.com/docs/guides/evm/precompiles/staking-v2. Accessed September 9, 2026.

[32] Bittensor. Stake from the EVM. https://www.bittensor.com/docs/guides/evm/stake-from-evm. Accessed September 9, 2026.

[33] Bittensor. Subnet mechanisms. https://www.bittensor.com/docs/guides/subnets#mechanisms. Accessed September 9, 2026.

[34] Slop Ninja Research. Public A models and private B generation. https://github.com/sam0x17/slopninja/blob/14d1ae1/docs/public-model-evaluation.md. Repository revision 14d1ae1; adopted design, implementation pending.

[35] Bittensor. Tempo: blocks per epoch. https://www.bittensor.com/docs/hyperparameters/tempo. Accessed September 9, 2026.

[36] Slop Ninja. Pangram repeatability probe. https://github.com/sam0x17/slopninja/blob/77b32b3/docs/pangram-repeatability.md. Repository revision 77b32b3.

[37] Bittensor. Mining. https://www.bittensor.com/docs/guides/mining. Accessed September 10, 2026.

[38] Bittensor. Subnet hyperparameters. https://www.bittensor.com/docs/hyperparameters. Accessed September 10, 2026.

[39] Bittensor. Emissions and Yuma Consensus. https://www.bittensor.com/docs/concepts/emissions. Accessed September 10, 2026.

[40] Bittensor. Precompile design and lifecycle. https://www.bittensor.com/docs/guides/evm/precompile-design. Accessed September 10, 2026.

[41] Internet Engineering Task Force. RFC 8949: Concise Binary Object Representation. https://www.rfc-editor.org/rfc/rfc8949.html. Accessed September 10, 2026.

[42] Bittensor. Chain consensus. https://www.bittensor.com/docs/concepts/chain-consensus. Accessed September 10, 2026.

[43] Pangram. AI detection API reference. https://docs.pangram.com/api-reference/ai-detection. Accessed September 9, 2026.

[44] Slop Ninja. Pangram public-result probe. https://github.com/sam0x17/slopninja/blob/77b32b3/docs/pangram-public-results.md. Repository revision 77b32b3.

[45] Bittensor. Validating: Commit-reveal. https://www.bittensor.com/docs/guides/validating#commit-reveal. Accessed September 9, 2026.

[46] Slop Ninja. Author projection study. https://github.com/sam0x17/slopninja/blob/77b32b3/docs/author-projection.md. Repository revision 77b32b3.

[47] Slop Ninja. Global Voices author transfer. https://github.com/sam0x17/slopninja/blob/77b32b3/docs/global-voices-transfer.md. Repository revision 77b32b3.

[48] Slop Ninja. Edit utility matrix and preservation counterexamples. https://github.com/sam0x17/slopninja/blob/77b32b3/docs/edit-utility-matrix.md. Repository revision 77b32b3.

[49] Slop Ninja. Encrypted receipt implementation. https://github.com/sam0x17/slopninja/blob/77b32b3/docs/receipt-envelope.md. Repository revision 77b32b3.

[50] Slop Ninja Research. Assignment and fulfillment gameability. https://github.com/sam0x17/slopninja/blob/2e5f7af/docs/assignment-gameability.md. Repository revision 2e5f7af; reproducible Rust experiment and results.

[51] Slop Ninja Research. Settlement gameability hypothesis. https://github.com/sam0x17/slopninja/blob/7ef473c/docs/settlement-gameability.md. Repository revision 7ef473c; synthetic experiment.

[52] Slop Ninja Research. Shared-pool settlement experiment. https://github.com/sam0x17/slopninja/blob/14d1ae1/docs/shared-pool-gameability.md. Repository revision 14d1ae1; synthetic experiment and pinned native math replay.